

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
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Originating Company	Epic Atlantic Limited
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R01	12-04-2022	For Review
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R04	03-11-2022	Re-issued for review after incorporating TCE comments

Revision Philosophy:

- All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPLFEED-GEN-RPT-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

The document, in consideration of relational design elements & concepts, forms the technical basis for the development of the facilities Project Specifications & tendering of contracts for the project's implementation.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
	FISAS Draft ESIA Report (November 2020)

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Project Information, Terms, Abbreviations, and Definitions

Project Information – General

1.0	Employer	BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED (BFPCCL)
2.0	PMC	TATA CONSULTING ENGINEERS LIMITED (TCE)
3.0	EPCC (CONTRACTOR)	CHINA TIANCHEN ENGINEERING CORPORATION (TCC)
4.0	Project Title	Gas Processing and Methanol Complex
5.0	Location	Akabel, Brass Island. The project site is about 85km South West of Port Harcourt, 70km South of Yenagoa and 90km West of Bonny Island.
5.1	State	Bayelsa State, Nigeria
5.2	Nearest Air Port and distance	Port Harcourt Airport (PHC) Bayelsa Airport (Under Construction)
5.3	Nearest Port	Port Harcourt (Onne)
5.4	Nearest Railway Station & distance	-
5.5	Highway and distance	Connected through water way only. Road connection does not exist.
5.6	Important Town and distance	Yenagoa, Nigeria & 67 km
5.7	Altitude Above Mean Sea Level	9 m
5.8	Atmospheric pressure	
	Average	1013 mbar
6.0	Ambient Air Temperatures	(Refer Note-5,6)
6.1	Maximum	38°C
6.2	Minimum	18°C
6.3	Average	23°C
6.4	Summer Design Wet Bulb Temp	28°C
6.5	Winter Design Wet Bulb Temp	15°C
6.6	MDMT	15°C

Project Information – General (Contd)

6.7	Swamp temperature	250C. (230C. to be used wherever a margin is required)
6.8	Ground temperature	25 - 27.50C.
6.9	Black-bulb temperature:	800C.
7.0	Relative Humidity	
7.1	Maximum	95 %
7.2	Minimum	51 %
7.3	Average	73 %
8.0	Rainfall Data	
8.1	Average annual rainfall	3800mm
8.2	Max daily rainfall:	180mm
8.3	Mean max. Hourly rainfall	100mm
9.0	Transport	
9.1	Name of the highway near which the plant is located	Connected through water way only. Road connection does not exist.
9.2	Access road	-
9.3	Railway (Gauge)	-
10.0	Wind Data	
10.1	Design Wind Velocity	35.6 m/sec (Note-1)
10.2	Wind Direction	Note-2
11.0	Seismic Data	
11.1	Earthquake Zone	Note-3
12.0	High Flood Level (HFL)	*
13.0	Natural Ground Level	*
14.0	Slope	*
15.0	Drainage	*
16.0	Design temperature for Electrical equipment	400C

Definition of Terms

Terms	Definitions
BFPCL	Brass Fertilizer & Petrochemical Limited.
SPDC	Shell Petroleum Development Company Nig. Limited
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)

Abbreviations

Abbreviation	Meaning
AASHTO	American Association of state Highway Transportation Officials
AGEF	Above Ground End Facilities
ALARP	As Low as Reasonably Practicable
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BOD	Basis Of Design
CP	Cathodic Protection
CPT	Cone Penetrometer Test
CTMS	Custody Transfer Metering Station
DB&B	Double Block and Bleed
DCQ	Daily Contractual Quantity

Abbreviations (contd.)

Abbreviation	Meaning
DEP	Design and Engineering Practice
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
FEED	Front End Engineering Design
FGS	Fire, Gas and Smoke detection systems
GPP	Gas Processing Plant
GTG	Gas Turbine Generators
HAZID	Hazard Identification
HAZOP	Hazard Operability
HEMP	Hazards and Effects Management Process
HFE	Human Factors Engineering
HPU	Hydraulic Power Unit
HRA	Health Risk Assessment
HSES	Health, Safety, Environment and Security
HSSE & SP	Health, Safety, Security, Environment and Social Performance
IEC	International Electro-technical Commission
ISPM	Industry Standard Process Modules
JV	Joint Venture
LCD	Liquid Crystal Display
LOI	Local Operator Interface
LRL	Lower Range Limit
LRV	Lower Range Value
LIRA	Logistics and Infrastructure Resource Assessment
MAC	Main Automation Contractor
NAG	Non-Associated Gas
NMS	Network Management Station
OPC	OLE for Process Control
OPPS	Overpressure Protection System

Abbreviations (contd.)

Abbreviation	Meaning
PCD	Process Control Domain
PDMS	Plant Design and Management System
PEPF	Package Engineering, Procurement, and Fabrication
PIG	Pipeline Inspection Gauge
PVC	Polyvinyl Chloride
RTR	Reliability Test Run
SCE	Safety critical elements
SI	International System Units
SIL	Safety Integrity Level
SRB	Sulphate Reducing Bacteria
SIS	Safety Instrumented System
SWB	Steel Wire Braiding
URV	Upper Range Value
USC	US Customary Units
XLPE	Cross Linked Polyethylene
TEG	Tri-ethylene Glycol
TQ	Top Quartile

1. Introduction

1.1. Background Information

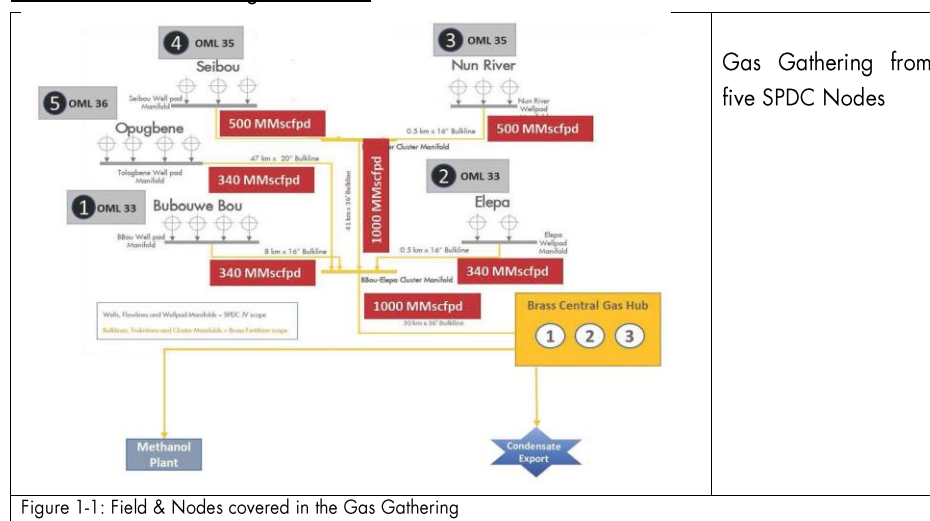
Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a greenfield methanol production facility in Akabel (Odioma), Brass LGA Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields, namely: Bubouwe Bou & Elepa, in OML 33, Nun River, Seibou, in OML 35, and Opugbene in OML 36.

The Project involves the development, construction, and operation of a 10,000 metric tonnes per day (MTPD) capacity methanol production plant comprising the following elements:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel, - as shown in Figures 1.1, 1.2, & 1.3.
- Product Export Lines and associated Offshore Single Buoy Mooring (SBM) for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) which are required for the operation of the processing plants.

Tata Consulting Engineers Limited (TCE) is the Project Management Consultant for the project.

The Nodal Gas Gathering Schematics



The Nodal Gas Gathering Schematics (contd.)

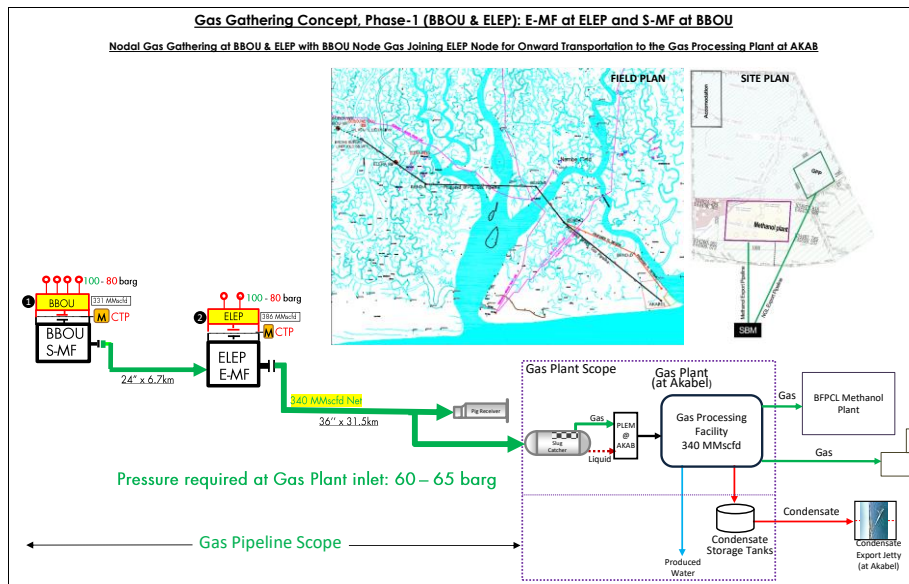


Figure 1-2: Gas Gathering Concept, Phase 1

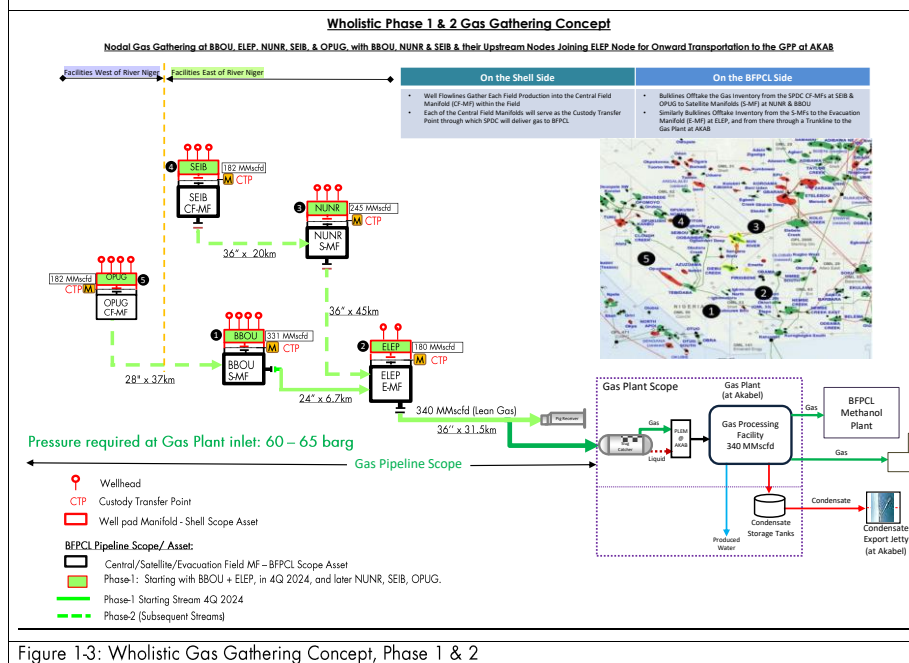


Figure 1-3: Wholistic Gas Gathering Concept, Phase 1 & 2

1.2. Purpose of Document

The purpose of this document is to provide the technical basis for the development of Project Philosophies and Specifications, including all the necessary information, design data, codes and standards required for the FEED of the gas pipelines and associated facilities. The design and construction of the pipelines shall fully comply with all relevant contractual requirements specified in the Scope of Work and Client's Standards.

The gas gathering will be in two phases, as shown in the schematics below:

- a. Phase 1 comprising gas gathering from Bubouwe Bou and Elepa fields, and
- b. Phase 2 comprising gathering from Nun River, Seibou, and Opugbene fields.

1.2.1. The Phase 1 Facilities

The phase 1 facilities (shown in Figure 1.1) are:

- BBOU – ELEP pipeline
- ELEP - AKAB Pipeline
- BBOU satellite Manifold
- ELEP evacuation Manifold, and
- The Pipeline End Manifold at AKAB.

These phase 1 facilities are the focus of the present FEED, however with consideration for provisions and retrofits that will accommodate the phase 2 facilities in a rather seamless manner.

The pipeline route proposed for this phase 1 has been surveyed, and the survey report issued for the FEED. The phase 2 pipelines routes are yet to be defined and surveyed.

1.2.2. The Phase 2 Facilities

The phase 2 facilities comprise NUNR, SEIB, and OPUG Manifolds with their pipelines connecting to phase 1 facilities as shown in Figure 1.2.

1.3. The Phase 1 Pipelines Route / Right of Way

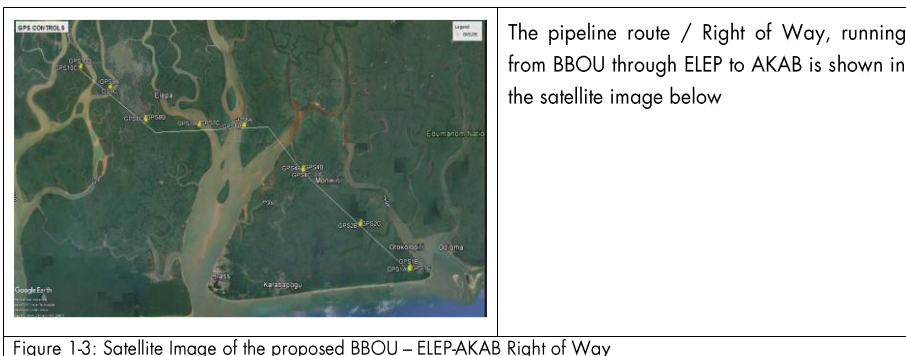


Figure 1-3: Satellite Image of the proposed BBOU – ELEP-AKAB Right of Way

1.4. Scope

The Facilities to be covered under project scope are:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from OPUG.
- Evacuation Manifold at ELEP complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from NUNR.
- Pipeline from BBOU to ELEP
- Pipeline from ELEP to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEP Manifold
- PIG Receivers at ELEP and AKAB
- Manifold control system also to be provided. Control Room required at each Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.
- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

The Scope of the FEED Engineering covers the provision of all necessary information, technical drawings, philosophies, specifications, design data, listing of codes and standards required for the detailed design and construction of the Multiphase pipeline system.

1.5. Design Battery limits

The battery limits for this Phase 1 FEED Project are:

1.5.1. BBOU Platform

- The terminal flanged ball valve on the manifold collecting well fluids from BBOU Wells.
- The flanged ball valve provided for the tie-in of Opugbene (OPUG) Pipeline on the manifold transporting well fluids to ELEP.

1.5.2. ELEP Platform

- The terminal flanged ball valve on the manifold collecting well fluids from ELEPA Wells.
- The flanged ball valve provided for the tie-in of NUN River Pipeline on the manifold transporting well fluids to AKAB.

1.5.3. Akabel Facility

- The Pipeline terminal flange at the inlet to the slug catcher at AKAB.
- The drain outlet flange on the pig receiver at AKAB.

2. Regulations, Codes And Standards

2.1. General

The regulations, codes, and standards to be applied shall always be the latest revision thereof and any amendment or supplement thereto in effect as at the contract date with contractor, manufacturer, or supplier. Design, fabrication, and installation work will be governed by the following regulations, codes, and standards.

2.2. Regulations

Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. Guidance on the statutory requirements for pipelines design, construction and operation should be obtained from the following documents:

2.2.1. Local Regulations

Regulation	Function
Petroleum Act, CAP P10, Laws of the Federation of Nigeria, LFN 2004, amended.	This is the principal statute regulating the oil industry. The following regulation relevant to this project is issued pursuant to the Act.
Mineral Oil (Safety) Regulations 1963, as amended.	These regulations, in so far as they apply to pipelines, require compliance with relevant IP, API and ASME codes and standards.
Oil Pipelines Act, 1956. CAP. 145 (Amended 1965, CAP. 338, LFN 1990)	Provides for licenses to be granted for the establishment and maintenance of pipelines incidental and supplementary to oilfields and oil mining, and for purposes ancillary to such pipelines.
Oil and Gas Pipelines Regulations, 1995	Details the regulations for the design, construction and maintenance of oil and gas pipelines.
Environmental Impact Assessment Act, CAP E12, LFN 2004.	Makes the conduct of an EIA mandatory prior to the development of any project or activity likely to have a significant effect on the environment
Federal Environment Protection Agency Act, CAP F13, LFN 2004.	Provides for the protection of the environment and sustainable development of Nigeria's natural resources.
Environmental Guidelines and Standards For The Petroleum Industry In Nigeria, 1991.	DPR (NUPRC) issued document providing guidelines on pollution abatement and relevant legislation.

Regulation	Function
Local Content Requirements	The provisions of the Nigerian Content Act apply to this project. The design includes strategies to comply with the Act and to maximize Nigerian local content throughout the life of the project. Exemptions / waivers have been sought where required.

2.3. Concept Engineering Referenced Documents

Document Number	Title
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001-3	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option
6064/PLRS/001a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.

2.4. Shell Design and Engineering Practices (DEPs)

Applicable DEPs, supplements and MESC SPE Codes are as referenced in various discipline dossiers. The design shall be based on DEP Feb 2019 (or latest, as advised by the EMPLOYER) some of which are listed below.

Document Number	Document Title
DEP 30.10.73.10-Gen	Cathodic Protection
DEP 30.10.73.31-Gen	Design of Cathodic Protection Systems for Onshore Buried Pipelines

Document Number	Document Title
DEP 30.48.00.31-Gen.	Painting and Coating of New Equipment
DEP 31.36.00.30-Gen	Pipeline Transportation Systems – Pipeline Valves (amendments/supplements to ISO 14313)
DEP 31.38.01.29-Gen	Pipe supports
DEP 31.40.00.10-Gen.	Pipeline Engineering
DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems
DEP 31.40.10.14-Gen	Pipeline Overpressure Protection
DEP 31.40.10.20-Gen	Flexible Composite Pipe System
DEP 31.40.20.37-Gen	Line pipe for critical service.
DEP 31.40.20.33-Gen	Line pipe induction bends (amendments/supplements to ISO 15590-1)
DEP 31.40.20.38-Gen	Selection and Design of Thermoplastic and Composite Pipeline Materials for Onshore Use
DEP 31.40.21.30-Gen.	Pipeline fittings (amendments/supplements to ISO 15590-2)
DEP 31.40.21.31-Gen	Pipeline Isolation Joints (amendments/supplements to ISO 15590-2)
DEP 31.40.21.33-Gen.	Pig signallars: Intrusive type
DEP 31.40.21.34-Gen.	Carbon and low alloy steel pipeline flanges for use in oil and gas operations (amendments/supplements to ISO 15590-3)
DEP 31.40.30.30-Gen	Concrete coating of line pipe
DEP 31.40.30.31-Gen	External polyethylene and polypropylene coating for line pipe
DEP 31.40.40.38-Gen	Hydrostatic pressure testing of new pipelines
DEP 31.40.50.30-Gen	Pre-commissioning of pipelines
DEP 31.40.60.11-Gen.	Pipeline leak detection

2.5. International Codes And Standard

Discipline	Document Number	Title
Civil	ACI 318	Structural Concrete Building Code
Civil	ASTM A185	Standard Specification for Steel Welded Wire Reinforcement, Plain, for Concrete
Civil	ASTM F552, 567, 626, 668, 1043	Chain Link Fencing
Civil	AWS D1.1	Structural Welding Code - Steel
Civil	EN 10080	Steel for the reinforcement of concrete
Civil	EN 1338	Concrete Paving Blocks. Requirements and test methods

Commented [AJ1]: 1) Contractor shall ensure all necessary checks, internal software checks, codal compliances to American codes as appropriate. American codes and standards shall be followed instead of British codes and Euro codes.

2) Contractor shall ensure thorough checking of all structural members Serviceability check and strength combinations to retain complete Structural integrity required for worst load case scenarios as per codal requirement.
3) Contractor to submit design calculation report for pipeline design and platform

Commented [H2]: All issues pertaining to codes and standards have resolved in a session by BFPCL/EAL/TCE

Discipline	Document Number	Title
Civil	EN 13670	Execution of Concrete Structures
Civil	EN 197	Cement. Composition, Specifications and Conformity Criteria for Common Cements
Civil	EN 1990	Eurocode — Basis of Structural Design
Civil	EN 1991	Structural Steel Loads & Design
Civil	EN 1992	Concrete Codes and Standards
Civil	EN 1993 Series	Equivalents in American & Canadian Codes accepted.
Civil	EN 1994 Series	Design of Composite Steel and Concrete Structures
Civil	EN 1997-1	Geotechnical design - Part 1: General rules
Civil	EN 206	Equivalents in American & Canadian Codes accepted.
Civil	EN ISO 3766	Construction Drawings - Simplified Representation of Concrete Reinforcement
Civil	ETAG-001 to 06	Guideline for European Technical Approval
Electrical	API 505	Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Zone 0, Zone 1, and Zone 2
Electrical	API 541	Form-wound Squirrel Cage Induction Motors - 375 kW (500 Horsepower) and Larger
Electrical	API 547	Form-wound Squirrel Cage Induction Motors – 250 Horsepower and Larger
Electrical	IEC 61000	Test Standard for Electrostatic Discharge (ESD) Immunity
Electrical	IEC 60034	Rotating electrical machines
Instrumentation & Control	ASME B40.100	Pressure Gauges and Gauge Attachments
Instrumentation & Control	ASME PTC 19.3	Thermowell Performance Test Code
Instrumentation & Control	BS EN 50288-7	Multi-Element Metallic Cables Used in Analogue and Digital Communication and Control. Sectional Specification for Instrumentation and Control Cables
Instrumentation & Control	IEC 60751	Industrial Platinum Resistance Thermometers and Platinum Temperature Sensors.
Instrumentation & Control	IEC 60079-14	Electrical Apparatus for Explosive Gas Atmospheres – Electrical Installation in Hazardous Areas (Other than mines).
Instrumentation & Control	IEC 60079-11	Explosive Atmospheres – Equipment Protection by Intrinsic Safety “i”.
Instrumentation & Control	IEC 60529	Degrees of Protection Provided by Enclosures (IP Code)

Discipline	Document Number	Title
Control		
Instrumentation & Control	IEC 61000-6-2	Generic Standards – Immunity Standard for Industrial Environments.
Instrumentation & Control	IEC 61511	Safety Instrumented Systems for the Process Industry.
Instrumentation & Control	ISO 12944-2	Paints and Varnishes – Corrosion Protection of Steel Structures by Protective Paint Systems.
Instrumentation & Control	API 521	Pressure Relieving and Depressuring Systems
Instrumentation & Control	API 526	Flanged Steel Pressure Relief Valves
Piping	ASME B31.3	Process Piping
Piping	ASME B1.2	Pipe Threads
Piping	ASME B16.11	Forged fittings, Socket Welding and Threaded
Piping	ASME B16.34	Valves – Flanged, Threaded and Welding Ends
Piping	ASME B16.5	Steel pipe flanges and flanged fittings
Piping	ASTM A 106	Spec for seamless CS pipe high temperature service
Piping	ASTM A 135	Spec for ERW Steel pipe
Piping	ASTM A 216	Specification for Steel castings, carbon suitable for High temperature service.
Piping	ASTM A 234	Spec for pipe fittings of wrought carbon steel and alloy steel for moderate & high temperature service
Piping	ASTM A105	Forgings, Carbon Steel, For Piping Components
Piping	ASME B16.9	Factory-made wrought steel butt welding fittings
Piping	ASME B36.10	Welded and seamless wrought steel pipe
Piping	ASME Section V	Non-destructive Examination.
Piping	ASTM A234	Standard specification for piping fittings of wrought carbon steel and alloy steel for moderate and elevated temperatures
Piping	ASTM A333	Standard specification for seamless and welded steel pipe for low temperature service
Piping	ASTM A420	Standard specification for piping fittings of wrought carbon steel and alloy steel for low temperature service
Piping	B16.20	Metallic Gaskets for Pipe Flanges: Ring-Joint, Spiral-Wound, and Jacketed
Piping	B16.21	Non-metallic Flat Gaskets for Pipe Flanges
Piping	MSS SP 25	Standard Marking systems for Valves, Fittings, Flanges and

Discipline	Document Number	Title
		Unions
Piping	MSS SP 58	Pipe Hangers and Supports –Materials, Design, Manufacture, Selection, Application, and Installation
Piping	BS EN 10204	Metallic Products – Types of Inspection Documents
Pipeline	API 1102	Steel Pipelines Crossing Railways and Highways
Pipeline	API 1104	Welding of Pipelines and Related Facilities
Pipeline	API 1109	Marking Liquid Petroleum Pipeline Facilities
Pipeline	API 5L	Specification for Line Pipe
Pipeline	API 6D	Dimensional standard for valves
Pipeline	ASME B 16.47 (series A)	Dimensional standard for large steel pipe flanges.
Pipeline	ASME B31.8	Gas Transmissions and Distribution Piping Systems.
Pipeline	ASME B46.1	Surface texture
Pipeline	ASME IX	Welding qualifications, ASME boiler and pressure vessel code
Pipeline	ASTM A 106	Spec for seamless CS pipe high temperature service
Pipeline	ASTM A 135	Spec for ERW Steel pipe
Pipeline	ASTM A 216	Specification for Steel castings, carbon suitable for High temperature service.
Pipeline	ASTM A 234	Spec for pipe fittings of wrought carbon steel and alloy steel for moderate & high temperature service
Pipeline	ASTM A105	Forgings, Carbon Steel, For Piping Components
Pipeline	ASTM A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
Pipeline	ASTM A694	Spec for Carbon & Alloy steel forgings for pipe flanges, fittings, valves and parts for high-pressure transmission service.
Pipeline	ASTM E92	Test Method for Vickers Hardness of Metallic Materials
Pipeline	ASTMA 134	Spec for electric fusion arc welded steel pipe.
Pipeline	MSS SP 75	Specification for High test Wrought Butt Welded Fittings
Pipeline	ISO 13623	Petroleum-and-Gas-Industries-Pipeline-Transportation-Systems
Pipeline	ISO 15590-1	Induction bends, fittings and flanges for pipeline transportation systems – Part 2: Induction Bends
Pipeline	ISO 15590-2	Induction bends, fittings and flanges for pipeline transportation systems – Part 2: Fittings
Pipeline	ISO 15590-3	Induction bends, fittings and flanges for pipeline

Discipline	Document Number	Title
		transportation systems – Part 2: Flanges
Geo-tech	ASTM D4254	Test Method for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density
Geo-tech	ASTM D2216	Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
Geo-tech	ASTM D1140	Standard Test Methods for Determining the Amount of Material Finer than 75- μ m (No. 200) Sieve in Soils by Washing
Geo-tech	ASTM D4318	Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
Geo-tech	ASTM D1557	Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft ³ (2,700 kN-m/m ³))
Geo-tech	ASTM D3080	Standard Test Method for Direct Shear Test of Soils Under Consolidated Drained Conditions

2.6. Conflict of Information

The edition of each document current at the date of enquiry shall apply, together with all published amendments. Where conflict occurs between the requirements of this Specification and referenced Codes and Standards, the Engineer shall notify EMPLOYER in writing immediately for resolution. In all cases, the most stringent shall apply. All deviations from the requirements of these Specifications, and the appendices and the referenced Codes and Standards shall be stated on a Technical Query (TQ) and issued to Employer for approval. In the absence of such a TQ, full compliance with the order of precedence below shall be assumed. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

3. General Design Data

3.1. Design Units of Measurement

Variable	Unit
barg	Bar Gauge (Pressure)
bara	Bar Absolute (Pressure)
bbl	Barrel (Oil Volume)
Bcf	Billion Cubic Feet
Bopd	Barrels of Oil Per Day
Bpd or B/D	Barrel Per Day
BS&W	Base Sediment and Water
Bwpd	Barrels of Water Per Day
cP	Centipoise (Unit of Viscosity)
F	Fahrenheit
C	Centigrade
km	Kilometre
m	Metre
Mbopd	Thousand Barrels of Oil per Day
Mbpd	Thousands Barrels Per Day
Mbwpd	Thousand Barrels of Water Per Day
MJ	Megajoule
mm	Millimetre
MMscfd	Million Standard Cubic Feet Per Day
MW	Mega-Watt
ppm	Parts Per Million
psig	Pounds Per Square Inch Gauge (Pressure)

3.2. Standard Process Conditions

Standard process conditions shall be taken as 15°C and 1.01325 bar.

3.3. Site and Environmental Conditions

The following site and environmental conditions have been extracted from Referencing paragraph 4.2.3 of Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited.

3.3.1. Air Temperature

Average maximum (daytime) temperature in the dry season months was about 32°C, and about 28°C in the rainy season. The minimum (nighttime) temperature level was fairly the same (23°C) throughout the year.

3.3.2. Weather

Two weather conditions are prevalent in the region:

Dry Season (Harmattan)

During Harmattan season (November – January, 2-3 months/year), the atmosphere is laden with extremely fine dust (particle size of 5 microns or less), which can penetrate equipment, block air filters, etc.

Wet Season

Heavy rainfall can be expected in the wet season which normally runs from the months of April through October. Lightning is very frequent especially during the rainy season and shall be specifically considered in the design of all relevant equipment.

3.3.3. Rainfall

Rainfall steadily increases from about 65mm in January to about 288mm in July after which it drops to about 245mm in August. This amount increases to a second maximum peak of about 331mm in September and then steadily drops to 47mm in December. The double maxima characteristics of Equatorial climate are very evident in the project area.

3.3.4. Relative Humidity

Parameters	Values
Mean	81%
Minimum	69%
Maximum	88%

3.3.5. Wind Speed and Direction

The monthly mean wind speed in the project area varies from 6.2km/h to 9.6km/h with a mean of 8.5km/h.

Commented [AJ3]: Site condition data from document number-TCE.11998A-ME-6177-DC-60001, Appendix-1 . EPIC provide what is the basis of this information of Site and Environment condition.

Commented [KY4]: The content of the paragraph gives the reference. It is from paragraph 4.2.3 of Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited. The document has no document number

The south-westerly wind is predominant (about 60% of the annual winds) and prevails during the wet season (March - October). The north-easterly wind which predominate during the dry season (November– February) and also brings the harmattan accounts to about 32% of the annual winds.

3.3.6. Seismic Zone

The project area is classified as seismic zone 0.

3.4. Design Life

All equipment for the project shall be designed based on a service life of 30 years.

4. Project Description

4.1. General Overview

The BFPCL and allied facilities' development will be constructed through four (4) major work packages – Gas Gathering Pipelines, Gas Processing Plant, the Methanol Plant and a Power Plant – with each package handled by different contractors. The work packages shall be managed to ensure seamless interfaces.

The scope for the FEED is the Gas Gathering Pipelines which will be installed in phases to match the phased business plan of BFPCL, starting with Phase 1.

Phase 1 comprises the Gas Supply bulk line from BBOU to ELEP, the trunk line from ELEP to AKAB and the remote manifolds at BBOU and ELEP

Phase 2 comprises pipeline systems and manifolds of NUNR, SEIB and OPUG.

However, while the Pipeline System will be designed for the GSPA DCQ of 340 MMscfd, opportunity will be taken to make the design robust to support the extended Business Plan of BFPCL for 1 MMMscfd gas hub. In this respect the pipeline system will be optimally sized to handle part of the 1 Bscfd, and with provision for tie-in of a loop line which the subject pipeline system will support to handle 1 Bscfd hub, as shown in the schematics below.

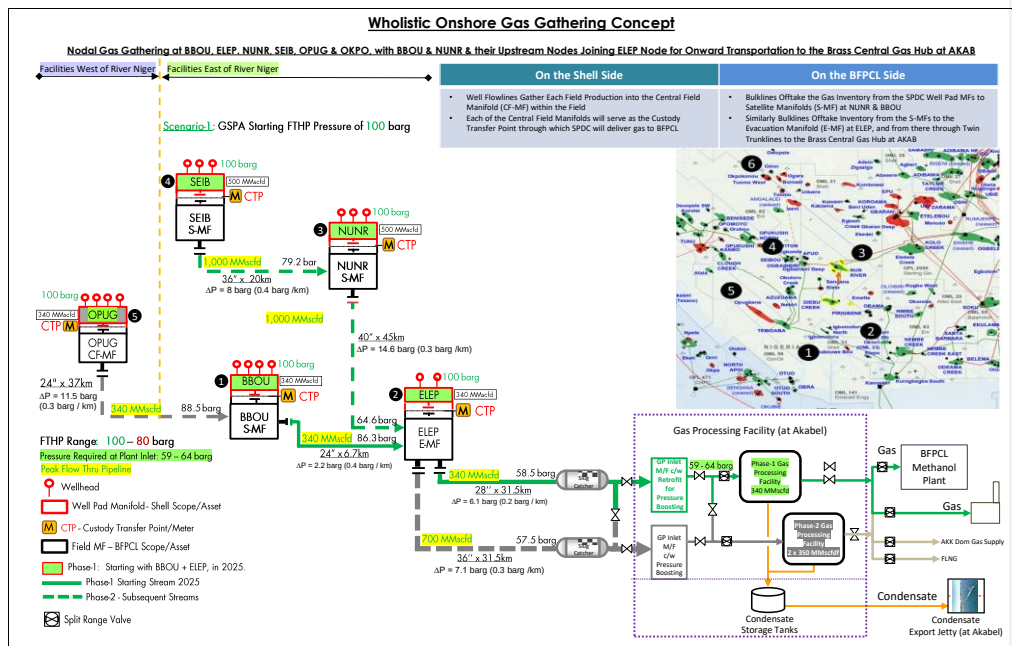


Figure 4-1: Wholistic Gas Gathering Concept, - for the 1 Bscfd Case

Materials and equipment to be used on the project shall, as much as possible, be locally and international proven-in-use and of standard sizes and materials. Availability of local experience and after sales support will be a major criterion in materials and equipment selection.

Materials and equipment in developmental stages and yet to be proven over a reasonable period of time shall not be used.

The Gas Gathering Pipelines and Associated Infrastructure project interfaces with SPDC (gas supplier) at the delivery point, and with the Gas Processing Plant at the inlet to the Slug Catcher. The delivery point is where custody transfer from SPDC to BFPCL takes place and is located at the field manifold. There will be some level of interaction and integration with the SPDC facilities and Gas Processing Plant and so standardization across the 3 Facilities, and in particular with the Methanol Complex, shall be pursued to allow for the benefits of commonality of spares, tools, maintenance routines, operations efficiency, support services and training.

4.1.1. Wind Speed and Direction

The monthly mean wind speed in the project area varies from 6.2km/h to 9.6km/h with a mean of 8.5km/h.

4.2. Concept Description

The project's concept is to gather raw wellhead gas and transport same to a Gas Processing Plant for processing as feedstock to Methanol Plant, and Power Plant.

The gas sources cut across five fields: Bubouwe Bou, Elepa, Nun River, Seibou and Opugbene, - in OMLs 33, 35, & 36, – with locations widely spread, and on both sides of Nun River.

The gas gathering & evacuation concept constitutes a mix of wellhead flowlines, field manifolds, bulklines, remote satellite manifolds, evacuation manifolds, and trunkline, optimally configured in terms of cost, as driven by manifolds layout and choice of evacuation manifold.

The overall development concept for the BFPCL project encompasses:

- a. Initial provision of manifolds, a bulkline and a trunk line from two startup wells at BBOU and ELEP in OML33 to convey produced fluids to the Gas processing Plant, including the provision of slug catcher at the Gas Plant end.
- b. Subsequent provision of bulk lines and manifolds to convey multiphase well streams from additional wells in four fields (NUNR, SEIB, OPUB and OKPO) located in OML's 35 and 36 to the Gas Plant.
- c. Installation of a gas treatment facility that is sized for 340 MMscfd net at GPP outlet.
- d. Installation of a gas powered power plant.
- e. The gas from the supply wells shall be gathered at the designated ELEP remote field manifolds and transported via bulk lines and trunklines to the Brass Gas Plant; where it will be treated and supplied to the methanol and power plants.

Condensate produced from the Gas Plant shall be transported, via a pipeline to an export terminal or to the Brass Terminal.

4.3. BOUBUWE (BBOU) Manifold Station

The station will comprise the following

- Field Manifold Headers – SPDC Wells & BFPCL Bulk
- Incoming OPUG-BBOU Bulk line Pig Receiver (future) & Outgoing BBOU-ELEP Bulk line Pig Launcher
- In-plot Piping System designed as per ASME B31.3
- Pig handling facilities
- RTU, Including control system and UPS for control system - These shall be in the Control Room/Field Auxiliary Room (FAR)

Power cable with back up solar power system

- Electrical Transformer & Electrical Room
- Chemical Injection System with storage for 2 – 4 weeks endurance
- Drain Pit or Saver Pit
- Guard Hut & Small Boarding
- Gantry & Accessories
- Telecommunication through fibre optic cable with VSAT System as back up
- High Security Fencing (e.g. Citadel, Heras)
- Navigation aids.
- Gas – over - oil actuator for valves required

4.4. 24" x 6.7km Pipeline System

The pipeline departing the BOUBUWE (BBOU) Manifold Station to ELEPA (ELEP) manifold station shall have the following appurtenances

- Sacrificial Anode / Bracelet Anode
- Fibre Optic Cable
- Power Cable (Optional)
- Cathodic Protection System

4.5. ELEPA (ELEP) Manifold Station:

The station will comprise the following

Commented [LMS5]: What is the provision being considered for instrument air required for valve actuation? Please mention

Commented [KY6]: The need for instrument air has been dropped because the actuated valves will be operated by a gas-over-oil actuator

Commented [LMS7]: Including control system, UPS for control system

Commented [KY8]: This comment is not clear on what's required to be done.

Commented [AJ9]: How much chemical storage is required which is kept at the manifold and share the logistic Plan.

Commented [KY10]: The size of the chemical injection storage will be provided in the datasheet for the Skid. This section of the BoD only lists the major components of the Manifold Station.

Commented [LMS11]: What is the provision being considered for instrument air required for valve actuation? Please mention

Commented [KY12]: The need for instrument air has been dropped because the actuated valves will be operated by a gas-over-oil actuator

- Field Manifold Headers – SPDC Wells & BFPCL Bulk
- Incoming BBOU-ELEP Bulk line Pig Receiver & Outgoing ELEP-AKAB Bulk line Pig Launcher
- Provision of space for the installation of Incoming NUN River-ELEP Bulk line Pig Receiver
- Pig handling facilities.
- In-plot Piping System designed as per ASME B31.3
- RTU including Control System- Control Room / Field Auxiliary Room (FAR)
- UPS for Control Room
- Telecommunication through fibre optic cable with VSAT System as back up
- Electrical Transformer & Electrical Room
- Chemical Injection System
- Drain Pit or Saver Pit
- Guard Hut & Small Boarding
- Gantry & Accessories
- High Security Fencing (e.g. Citadel, Harass)
- Navigational aids
- Venting system and Vent Stack
- Gas – over - oil actuator for valves required

Commented [AJ13]: What about Future PIG Receiver (NUNR-ELEP Link).

Commented [KY14]: A bullet point has been added to reflect this.

Commented [LMS15]: Including control system, UPS for control system

Commented [KY16]: Comment is not clear on what is required to be done

Commented [AJ17]: How much chemical storage is required which is kept at the manifold and share the logistic Plan.

Commented [KY18]: The size of the chemical injection storage will be provided in the datasheet for the Skid. This section of the BoD only lists the major components of the Manifold Station.

4.6. 36" x 31.5km Pipeline System

The pipeline departing Elepa (ELEP) manifold station Akabel Plant shall have the following appurtenances.

- Pig Launcher (at BBOU) / Receiver (at ELEP)
- Sacrificial Anode / Bracelet Anode
- Fibre Optic Cable
- Power Cable (Optional)
- CP System.
- Earthing System (at AKAB)

4.7. AKABEL Station (Complex)

- Incoming ELEP-AKAB Bulk line Pig Receiver
- Sludge catcher (at AKAB)
- In-plot Piping System designed as per ASME B31.3
- Venting system and Vent Stack

4.8. Design and Operating Philosophy

The following issues of practical importance have been embraced in the design & operating philosophy of the Pipeline System:

1. Design that will be in compliance with all relevant contractual requirements specified in the

Scope of Work and Client's Standards.

2. Design integration between SPDC and BFPCL in respect of Pipelines and Wellhead Manifold Platforms.
3. Design that will be environmentally friendly, in respect of containment of inventory, with zero spill to the environment.
4. Sizing of the pipeline segments to deliver the gas to the Gas Plant at pressure of 60 – 65 barg required at the gas plant intake, or with a rational economic assistance of pressure boosting that effect.
5. Selection of suitable materials that will withstand the fluid flow conditions & fluid properties, and the external environment, in respect material deterioration.
6. Specification of wall thickness of line pipe and piping class capable of withstanding design pressure.
7. Specification of corrosion allowance, and corrosion mitigation system that will sustain system's integrity over life cycle.
8. Placement of the facilities at a height that is reasonably above flood level that will ever be experience in the area, - drawing lessons from the 2012 rare flood experience where many oil and gas facilities in the Niger Delta Region were submerged, leading to inoperability of the facilities and subsequent extra-ordinary revamp to bring back the facilities back to operation.

5. HSSE and Social Performance (SP)

The following sections contain the Basis for Design project requirements for Health, Safety, Security, Environment and Social Performance for the Brass Fertilizer Gas Gathering & Associated Infrastructure Project.

5.1. Hazard & Risk Management

The hazard and risk management process will be followed to demonstrate that the risks to people, assets, reputation, the environment and social matters from the hazards are reduced to as low as reasonably practicable (ALARP). This is achieved by complying with Nigeria regulations and relevant Industry Standards.

5.1.1. Nigeria Legislation

The HSSE & SP philosophy for Brass Fertilizer Gas gathering Pipeline & Associated Infrastructure Project will set out the minimum Health, Safety, Security and Environmental (HSSE) and Social Performance (SP) requirements associated with the Project. The aim is to comply with Nigeria Regulations (HSE legislation), International Standards and Project Specific Requirements. The aim of the philosophy is to ensure that all HSSE & SP risks are reduced to as low as reasonably practicable, using the following risk reduction hierarchy approach.

- Hazard elimination
- Reducing the probability of hazardous events occurring
- Minimising the escalation risk should a hazardous event occur

5.1.2. ALARP

The general approach is to identify and assess the potential hazards; develop hazard management strategies for minimising the risk; assess the risk to conclude if the risks are adequately minimised or if further risk reduction measures are justified. The ALARP procedure requires that the following three questions are answered to establish if the risks are reduced to ALARP:

- Have all applicable laws and regulations been complied with?
- Has all relevant Good Practice been followed?
- Are there any more Reasonably Practicable measures that reduce the Risks further?

5.2. Health

5.2.1. Health Risk Assessment

HRA as an application of HEMP to the potential health hazards, shall be carried out to identify suitable control measure for all health hazards involved throughout the project activities using the following hierarchy of control:

- Eliminate the health hazard;
- Substitute the health hazard with a less harmful type or form;
- Use engineering controls;
- Use operational/ procedural controls; and
- Use personal protective equipment.

5.2.2. Medical

The project shall provide adequate medical facilities at the remote platforms that are capable to sustain and stabilise injured staff until Medivac arrives. The following guidelines shall apply:

- First aid treatment, including defibrillation, by a Designated First Aider within four minutes
- Assessment and stabilisation by a medical Emergency professional within one hour;
- Admission to and care at the nearest Local Hospital within four hours; and
- Referral to an appropriate Specialist Hospital (casualty specific).

When the above response times or requirements cannot be met, perform a Risk Assessment and provide medical Emergency Response Risk mitigation measures to ensure that the Risks are ALARP

5.3. Human Factors Engineering (HFE)

A Human Factors Engineering (HFE) Screening shall be conducted during the FEED. A strategy and an action plan shall be developed for integrating HFE as part of the phases of the project.

5.4. Safety

The objectives of the design of the gathering pipeline & infrastructure are to create an inherently safe design and whilst by demonstrating that risks to personnel are tolerable and reduced to as low as reasonably practicable (ALARP). During the FEED and subsequent project phases, HEMP shall be applied to the project and involves the following:

- Identifying and assessing all hazards.
- Identifying the Major Accident Hazards that require a more detailed assessment.
- Quantifying the consequences of the potential major accident hazards.
- Developing hazard management strategies to prevent, control, detect and mitigate the hazards based upon recognised good practice.

Philosophies shall be developed covering the following:

- Fire and explosion management philosophy covering design principles for active and/or passive fire protection and explosion protection
- Emergency response philosophy covering design principles for Medical Emergency Response, Escape and Evacuation Routes and Lifesaving Appliances & Personnel Protective Equipment.

5.4.1. Hazards and Effects Management Process (HEMP)

The hazards associated with the project will be managed using the Hazards and Effects Management Process (HEMP). This process provides a demonstration to stakeholders that major hazards (to people, assets, environment and reputation) have been identified and their risks reduced to a level that is ALARP. HEMP can be summarized as the following steps that will be implemented by the Project during the FEED and subsequent phases of the project:

- Identify credible and foreseeable HSSE & SP hazards and potential effects
- Assess the identified risks using the Project Risk Assessment Matrix
- Record identified potential hazards and their possible consequential impacts in the Hazards & Effects Register for the Project
- Establish risk-reduction measures through the adoption of the hierarchy of hazard management, including emergency preparedness and recovery measures
- Produce and maintain a documented demonstration that major HSSE & SP risks have been reduced to be ALARP.

An action tracking system shall be in place to manage all raised actions from the HEMP studies. All actions raised are tracked until formal close out.

5.4.2. HSSE & SP Case

The HSSE Case demonstrates that during the development of the project, there has been a systematic application of the Hazards and Effects Management Process (HEMP). It also aims at demonstrating that identified risk has been actively and systematically reduced to a level which is As Low As Reasonably Practicable (ALARP).

The details of how the major hazards identified are reduced to ALARP by implementing hazard mitigation measures in the FEED phase and defining further mitigation steps for the follow-on works, and documenting same in Design HSE Case. The Design HSE Case will be further developed in the subsequent project phases, to continue to identify ways to reduce risk to ALARP, until Operations HSE Case is developed as part of handover of the project to the Asset Owners.

5.4.3. Technical Integrity Verification

Technical integrity verification as the process by which the HEMP is put into practice for effective barriers will be followed. The essential elements of these barriers (Safety Critical Elements (SCEs) must be designed constructed and maintained in accordance with specified performance standards. This requires effective design, quality assurance, maintenance, verification and deviation processes.

At the FEED, SCEs will be identified and performance standards established. Technical Integrity Verification (TIV) shall be used to demonstrate that the design, procurement and construction processes have been properly applied to the project, to provide assurance of Technical Integrity

of the Asset.

5.5. Security

A Security hazards in design HAZID workshop shall be carried out during the FEED to identify security hazards early in design to ensure engineering solutions could be used to mitigate the hazards. The Security hazard in Design report shall form the basis of security risk reduction measures which will be detailed in the Security Plan. The Security Plan will be updated as project progresses and ensuring it is fit for purpose for the project as it develops.

5.6. Environment

The project facilities shall be designed and operated to ISO-14001 certification standards. This section defines the key parameters that form the environmental basis of design. The aim is to demonstrate that environmental risks are tolerable and impacts reduced to as low as reasonably practicable (ALARP). The over-arching philosophies for managing impacts to the environment include

- Energy use, energy optimisation and minimisation of atmospheric emissions shall be evaluated as part of the decision making process when assessing options for the main energy consumption items.
- Key design decisions regarding venting shall aim to minimise atmospheric emissions associated with the project facilities
- Production of non-gaseous waste shall be avoided, where possible, by using clean technologies or using waste minimisation/waste recovery technologies
- Generation of wastes and discharges of wastewater shall be minimised.

The environmental basis of design shall be reviewed and updated as the project matures to reflect any design decisions that might have an impact on the environmental performance. Relevant findings of future lessons learned reviews, as well as recommendations received through the update of the Environmental Impact Assessment report, shall also be taken into account.

5.6.1. Atmospheric Emissions

A Greenhouse Gas (GHG) and Energy Efficiency Plan shall be prepared during the FEED, to identify, quantify potential GHG emissions and propose project opportunities to improve energy efficiency. The GHG & EE plan will propose measures to ensure atmospheric emissions are minimized. Energy use and equipment selection should be optimised for normal and maximum operations and should take into account the full operational lifecycle of the facility including turndown/late life. Emissions of SO_x and NO_x from point sources in all Project installations should comply with local legal requirements.

5.6.2. Venting

There shall be no continuous venting of hydrocarbons. A Greenhouse Gas (GHG) and Energy

Efficiency Plan shall be developed during the FEED to identify, quantify potential GHG emissions and propose project opportunities to improve energy efficiency. Further project opportunities to reduce venting shall be identified. Design shall incorporate minimisation of potential leak sources to avoid fugitive emission and provision of a closed drainage system for the hydrocarbon systems, operational controls shall be established and maintained to ensure VOC emissions are reduced to As Low As Reasonably Practicable

5.6.3. Chemicals

Design shall:

- Minimise the use of chemicals
- Not use chemicals with a substitution warning and
- Ensure an auditable chemical assessment and selection process is used to demonstrate chemicals with the least environmental impact are adopted, where practicable.
- Ensure fit for purpose bunding is provided around tanks containing chemicals, lube oils, diesel, etc.

5.6.4. Spill Prevention

To minimise the potential for and to reduce risks of leaks and spills at source, the design shall ensure that:

- Leak paths are minimised.
- Bunded areas are designed fit for purpose and adequate space allowance is provided.
- Diesel tanks and re-fuelling location requirements are assessed.

5.6.5. Waste

The design shall ensure sufficient space allowance for Waste containers to allow adequate segregation and storage of Waste on the sites. Design shall: Incorporate controls to reduce Waste generation into Procedures and working practices. Where appropriate: - Identify opportunities to reuse Waste for the same or alternative applications, including in other industries, or allow for provision to return unused materials to suppliers. - Identify recycling and recovery opportunities for Waste material.

5.6.6. Impact Assessment

There shall be Project EIA report, subjected to the EIA iterative process and approvals. Several formal engagements shall be undertaken with DPR, Federal Ministry of Environment (FMEnv) regarding impact assessment scoping and supporting studies to ensure that complete update of the EIA early in FEED phase and subsequent update in the later phases of the project.

5.7. Social Performance

Host communities shall be identified and Social Performance workshop(s) shall be undertaken to

form the basis of the project specific social performance plan which will be developed for the project detailing how the project intends to manage its project foot print (social impacts) and incorporate the local requirement on Social Performance.

5.8. HSSE-SP Regulations & Standards

5.8.1. Nigerian Regulations & Standards

- Environmental Guidelines And Standards for the Petroleum Industry in Nigeria (EGASPIN 2018)
- Minerals Oil & Safety regulation MOSR 1997
- DPR Safety Case Guidelines for Oil & Gas Facilities in Nigeria (October 2020)

5.8.2. International Standards

- IP Code – Part 15: Hazardous Area Classification Guide
- IEC 61508 - Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems

5.8.3. Shell DEPs

- DEP 80.47.10.30-Gen: Assessment of Fire Safety of Onshore Installations
- DEP 80.47.10.31-Gen: Active Fire Protection Systems and Equipment for Onshore Facilities
- DEP 80.47.10.32-Gen: Portable and Mobile Equipment for Fire-Fighting

6. Process Basis of Design

The pipeline shall be designed for multiphase flow due to the substantial amount of water and condensate in the constituent gas.

6.1. Project Location & Operational Conditions

The project location plan & operational conditions are summarized in the table below.

1.	Location	OML 33 & 35 - with the fields widely spread & on both sides of Nun River
2.	Target Fields Phase 1 starting fields (Subject of this BOD)	BBOU & ELEP in OML 33 - East of Nun River
3.	Target Fields Phase 1 subsequent fields	NUNR in OML 35, - East of Nun River SEIB in OML 35 - West of Nun River
4.	Target Fields Phase 2 fields	OPUG in OML 36 and OKPO in OML 35, - West of Nun River
5.	Desired Offtake Rate / GSPA Contractual Rate	340 MMscfd for 25 years / TBA
6.	Gas Offtake Points	SPDC Field Manifold Headers at the target fields, in phases starting with BBOU & ELEP, followed by NUNR, SEIB and OPUG
7.	Feed Stream Inlet Pressure to BFPC P/L System	80 barg – 100 barg
8.	Feed Stream Outlet Pressure from BFPC P/L System to Gas Plant	60 barg – 65 barg
9.	Gas Plant Processed Gas Outlet Pressure	41 barg – 49 barg
10.	Condensate rate	11 Mbcpd in year 1 to 34.4 Mbcpd in year 18.
11.	Water Production	Starting at 1.8 Mbwpd in year 1 and increasing to about 16.5 Mbwpd in year 14.

Commented [AJ19]: Offtake should be commensurate with 340MMscfd Treated Gas at GPP Outlet

Commented [KY20]: Comment is not clear on what is to be done. Kindly clarify further.

Commented [AJ21]: 60-65 barg

6.2. Production Forecast & Fluid Characteristics

Notional Fields Forecasts of Well Head Gas (WHG) Supply Required For Export Of 340 MMSCFPD Of Lean Gas To Methanol Plant For 20 Years With Elepa Development Accelerated to Year 1 OSD with Bubouwe Bou

6.2.1. BUBOUWE BOU

Condensate-Gas Ratio (CGR) is 5.56 BBL/MMscf.

Time (Year)	Gas (MMSCF/D)	Condensate (BCPD)	Water (BWPD)
1	181	944	907
2	181	878	907
3	181	788	907
4	181	742	7855
5	181	707	7547
6	215	840	8491
7	305	1217	9002
8	327	1323	8193
9	331	1338	8038
10	67	239	2397
11	21	59	1031
12	16	42	784
13	2	4	85
14	0	0	0
15	0	0	0
16	0	0	0
17	0	0	0
18	0	0	0
19	0	0	0
20	0	0	0
TCQ	800	3	21
	(BSCF)	(MMBBL)	(MMBBL)

Commented [AJ22]: CGR is 4.35 as per the latest wellhead forecast. Please check.

Commented [H23]: 5.56 Confirmed. Value as received from SPDC

6.2.2. ELEPA

Condensate-Gas Ratio (CGR) is 60.64 BBL/MMscf.

Time (Year)	Gas (MMSCF/D)	Condensate (BCPD)	Water (BWPD)
1	180	10088	900
2	180	10088	900
3	180	10088	900
4	180	10088	7980
5	180	10088	7666
6	145	8146	4902
7	52	2911	2597
8	29	1630	1454
9	25	1386	1236
10	19	1076	960
11	11	628	561
12	9	478	426
13	1	52	46
14	0	0	0
15	0	0	0
16	0	0	0
17	0	0	0
18	0	0	0
19	0	0	0
20	0	0	0
TCQ	435	24	11
	(BSCF)	(MMBBL)	(MMBBL)

6.2.3. NUN RIVER

Condensate-Gas Ratio (CGR) is 67.57 BBL/MMscf.

Time (Year)	Gas (MMSCF/D)	Condensate (BCPD)	Water (BWPD)
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0
6	0	0	0
7	0	0	0
8	0	0	0
9	0	0	0
10	244	14464	1221
11	245	14583	1225
12	245	14583	1225
13	245	14583	10844
14	245	14583	10417
15	196	10914	6653
16	71	4159	3528
17	40	2329	1976
18	34	1980	1680
19	26	1538	1304
20	15	898	762
TCQ	586	35	15
	(BSCF)	(MMBBL)	(MMBBL)

6.2.4. SEIB+OPUG+ADD**

Condensate-Gas Ratio (CGR) is 65.63 BBL/MMscf.

Time (Year)	Gas (MMSCF/D)	Condensate (BCPD)	Water (BWPD)
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0
6	0	0	0
7	0	0	0
8	0	0	0
9	0	0	0
10	52	3049	259
11	108	6367	541
12	116	6831	580
13	139	8161	2857
14	141	8322	6096
15	188	11082	7496
16	308	18164	7717
17	341	24229	6297
18	348	32453	3334
19	354	31623	3116
20	363	31213	5551
TCQ	898	66	16
	(BSCF)	(MMBBL)	(MMBBL)

6.2.5. TOTAL SUPPLY PORTFOLIO

	Total WHG Supply*		CGR (BBL/MMscf) = 64.14
Time	GAS	CONDENSATE	WATER
year	MMSCF/D	BCPD	BWPD
Year	Gas	Condensate	Water
1	361	11032	1807
2	361	10965	1807
3	361	10876	1807

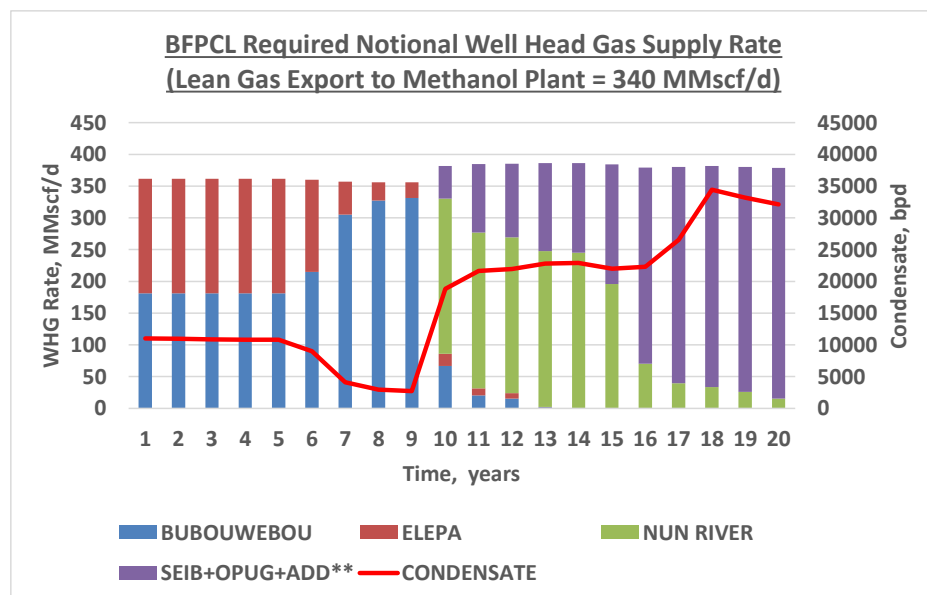
	Total WHG Supply*		CGR (BBL/MMscf) = 64.14
4	361	10830	15835
5	361	10794	15213
6	360	8985	13393
7	357	4128	11599
8	356	2953	9647
9	356	2724	9275
10	382	18828	4837
11	385	21637	3357
12	385	21933	3015
13	386	22799	13832
14	386	22905	16513
15	384	21996	14149
16	379	22323	11246
17	380	26558	8272
18	382	34433	5014
19	380	33161	4421
20	379	32111	6313
	2719	129	63
	(BSCF)	(MMBBL)	(MMBBL)

Notes

- WHG Supply* on Operating Day basis AND NOT Calendar day basis
- SEIB+OPUG+ADD** is subject to SPDC providing requested information
- Delivery Pressure range at Custody Transfer Point: 80 - 100 bar
- Delivery Temperature range at Custody Transfer Point: 50 - 95 degC

6.2.6. BFPCL Required Notional Well Head Gas Supply Rate

(Lean Gas Export to Methanol Plant = 340 MMscf/d)



6.2.7. Gas Compositions from Nominated Fields

6.2.7.1. Field Compositions

Field 1: Bubouwe Bou		Field 2: Elepa		Field 3: Nun River H4.0	
Analogue					
Component	Mol.%	Component	Mol.%	Component	Mol.%
N ₂	0.07	N ₂	0	N ₂	NA
CO ₂	0.59	CO ₂	2.22	CO ₂	0.77
C ₁	92.92	C ₁	81.66	C ₁	72.83
C ₂	3.16	C ₂	6.13	C ₂	7.68
C ₃	1.01	C ₃	2.59	C ₃	4.85
iC ₄	0.61	iC ₄	0.75	iC ₄	1.46
nC ₄	0.44	nC ₄	0.81	nC ₄	1.97
iC ₅	0.25	iC ₅	0.45	iC ₅	0.95
nC ₅	0.19	nC ₅	0.33	nC ₅	0.76
C ₆	0.23	C ₆	0.65	C ₆	1.03
C ₇	0.19	C ₇	0.9	C ₇	1.41
C ₈	0.04	C ₈	0.87	C ₈	1.61
C ₉	0.03	C ₉	0.38	C ₉	0.9

Field 1: Bubouwe Bou Analogue		Field 2: Elepa		Field 3: Nun River H4.0	
C ₁₀	0.02	C ₁₀	0.3	C ₁₀	0.65
C ₁₁	0.03	C ₁₁	0.23	C ₁₁	0.47
C ₁₂₊	0.22	C ₁₂	0.22	C ₁₂	0.37
C ₁₃	NA	C ₁₃	0.23	C ₁₃	0.34
C ₁₄	NA	C ₁₄	0.22	C ₁₄	0.3
C ₁₅	NA	C ₁₅	0.19	C ₁₅	0.26
C ₁₆	NA	C ₁₆	0.15	C ₁₆	0.21
C ₁₇	NA	C ₁₇	0.12	C ₁₇	0.16
C ₁₈	NA	C ₁₈	0.11	C ₁₈	0.14
C ₁₉	NA	C ₁₉	0.06	C ₁₉	0.08
C ₂₀₊	NA	C ₂₀₊	0.43	C ₂₀₊	0.8
TOTAL	100	TOTAL	100	TOTAL	100
Flash CGR	7	Flash CGR	59.7	Flash CGR	161.7

6.2.7.2. Portfolio Composition

Range for Portfolio	Lean Sample		Rich Sample
Component	Mol. %		Mol. %
N ₂	0.07		NA
CO ₂	0.59		0.77
C ₁	92.92		72.83
C ₂	3.16		7.68
C ₃	1.01		4.85
iC ₄	0.61		1.46
nC ₄	0.44		1.97
iC ₅	0.25		0.95
nC ₅	0.19		0.76
C ₆	0.23		1.03
C ₇	0.19		1.41
C ₈	0.04		1.61
C ₉	0.03		0.9
C ₁₀	0.02		0.65
C ₁₁	0.03		0.47

Range for Portfolio	Lean Sample		Rich Sample
C ₁₂₊	0.22		0.37
C ₁₃	NA		0.34
C ₁₄	NA		0.3
C ₁₅	NA		0.26
C ₁₆	NA		0.21
C ₁₇	NA		0.16
C ₁₈	NA		0.14
C ₁₉	NA		0.08
C ₂₀₊	NA		0.8
TOTAL	100		100

Notes

- STB - Stock Tank Barrel
- MMscf - Million standard cubic feet

6.2.7.3. Well Compositions

6.2.7.3.1.BBOU D3

As used in simulation model - Tuned PVTsim Model using Bonn-31ST PVT as analogue.

Component	Mol %	Molecular Weight	Liquid Density (g/cc)
N ₂	0.070	28.014	
CO ₂	0.590	44.010	
C ₁	92.922	16.043	
C ₂	3.160	30.070	
C ₃	1.010	44.097	
iC ₄	0.610	58.124	
nC ₄	0.440	58.124	
iC ₅	0.250	72.151	
nC ₅	0.190	72.151	
C ₆	0.230	86.178	0.664
C ₇	0.190	96.000	0.738
C ₈	0.040	107.000	0.765
C ₉	0.030	121.000	0.781
C _{10-C11}	0.050	141.800	0.794
C ₁₂	0.068	161.000	0.800

Component	Mol %	Molecular Weight	Liquid Density (g/cc)
C13	0.047	175.000	0.805
C14	0.032	190.000	0.808
C15	0.022	206.000	0.812
C16	0.015	222.000	0.815
C17	0.010	237.000	0.818
C18	0.007	251.000	0.821
C19-C44	0.016	293.067	0.830
Total	100.000		

Bonny 31ST Reservoir Fluid Composition				
No	Component	Flash Gas	Flash Oil	Reservoir Fluid
		[mol %]	[mol %]	[mol %]
1	N ₂	0.07	0	0.07
2	CO ₂	0.59	0	0.59
3	C ₁	93.26	0	92.92
4	C ₂	3.17	0.01	3.16
5	C ₃	1.01	0.03	1.01
6	i-C ₄	0.61	0.07	0.61
7	n-C ₄	0.44	0.07	0.44
8	i-C ₅	0.25	0.16	0.25
9	n-C ₅	0.19	0.16	0.19
10	C ₆	0.22	1.88	0.23
11	C ₇	0.17	6.21	0.19
12	C ₈	0.02	6.61	0.04
13	C ₉	0	7.13	0.03
14	C ₁₀	0	5.7	0.02
15	C ₁₁	0	7.83	0.03
16	C ₁₂₊	0	64.14	0.22

Total		100	100	100
Density @ 60F	[g/cm ³]		0.811	
MW	[g/mol]	17.97	165.61	18.47
GOR	[Scf/Stb]			180,011.90
Gravity	[air=1]	0.62		

C ₇₊	Mole %	0.45	97.62	0.53
C ₇₊ MW	[g/mol]	100.89	167.69	143.17
C ₁₂₊ MW	[g/mol]	-	191.6	191.6
CGR (bbls/mm scf)				5.56

Commented [AJ24]: Why CGR is different in PVT data and in well head Forecast data. Please check

Commented [H25]: Data as received from SPDC

6.2.7.3.2. Elepa U-4

Component	Res Fluid	Liquid		Gas	
	(z)		(x)		(y)
Totals	mol%		mol%		mol%
N2	0.0000		0.00		0.00
CO2	2.2200		0.00		2.30
H2S	0.0000		0.00		0.00
c1	81.6600		0.00		84.63
c2	6.1300		0.03		6.35
c3	2.5900		0.27		2.67
ic4	0.7500		0.37		0.76
nc4	0.8100		0.54		0.82
ic5	0.4500		0.87		0.43
nc5	0.3300		0.68		0.32
c6	0.6500		2.26		0.59
c7	0.9000		6.75		0.69
c8	0.8700		12.76		0.44
c9	0.3800		10.87		0.00
c10	0.3000		8.62		0.00
c11	0.2300		6.64		0.00
c12	0.2200		6.15		0.00
c13	0.2300		6.45		0.00
c14	0.2200		6.16		0.00
c15	0.1900		5.51		0.00
c16	0.1500		4.37		0.00
c17	0.1200		3.36		0.00
c18	0.1100		3.02		0.00
c19	0.0600		1.62		0.00
c20+	0.4300		12.70		0.00

Component	Res Fluid	Liquid		Gas	
Totals	100.00		100.00		100.00

6.2.7.3.3.Elepa U-7

Component	Res Fluid	Liquid		Gas	
	(z)		(x)		(y)
Totals	mol%		mol%		mol%
co2	2.1400		0.00		2.16
N	0.0000		0.00		0.00
c1	86.6700		0.00		87.30
c2	6.2100		0.37		6.25
c3	2.0500		0.24		2.06
ic4	0.5200		0.16		0.52
nc4	0.4900		0.26		0.49
ic5	0.2500		0.39		0.41
nc5	0.1600		0.36		
c6	0.2800		1.88		0.27
c7	0.3700		5.97		0.33
c8	0.3100		13.52		0.21
c9	0.1000		13.33		0.00
c10	0.0700		10.03		0.00
c11	0.0500		7.46		0.00
c12	0.0500		6.75		0.00
c13	0.0500		6.70		0.00
c14	0.0400		6.01		0.00
c15	0.0400		4.99		0.00
c16	0.0300		4.02		0.00
c17	0.0200		3.00		0.00
c18	0.0200		2.84		0.00
c19	0.0200		2.23		0.00
c20+	0.0700		9.49		0.00
Totals	100.01		100.00		100.00

6.3. Flow Assurance

A preliminary hydraulics study was carried out during the concept selection stage. A steady state hydraulic analysis shall be executed in the course of FEED phase to determine the following:

- Confirmation of the preliminary line sizes
- Optimise design / operating pressures for the pipelines.
- Establish expected flow regimes
- Liquid hold-ups, slug generated volumes and turndown capacities, and the impacts on slug management
- Evaluate sensitivities to line pack, flow rate ramp up / turn down and pigging operations
- Evaluate the impact of installation of new pipeline on the operability/maintainability of existing facilities.

The result is detailed in the Pipeline Hydraulic Study Report.

Steady State hydraulic analysis of the Multiphase Pipeline shall be carried out using PIPESIM Software. The results of the Pipeline hydraulic analysis shall be considered in the design of the pipeline system.

A transient hydraulic analysis shall be carried out during the FEED, and shall be incorporated into the startup, shutdown and normal operations narratives of the Pipelines Operations and Maintenance Manual.

6.4. Materials for Construction

Line pipe materials for construction of the new pipeline sections shall be carbon steel API 5L Gr X70. with 3LPE and concrete coatings. The selection of materials shall consider the maximum and minimum process conditions, the high strain capacity requirements, all-seasons flooded swamp terrain, the extremes of ambient temperatures and all other conditions that are likely to affect the performance of the material. Particular attention shall be given to minimize the potential for brittle fracture of materials due to depressurization and shall be addressed in a pipeline depressurization / blowdown study carried out during detailed design. The pipeline shall be protected with chemical injection due to the CO₂ content of the gas.

6.5. Pipeline Design Parameters.

Design Condition	Value
Design Pressure	110 barg
Operating Pressure	100 barg
Min/Max. Design Temperature	12.7/80°C
Operating Temperature	50°C

Pipeline Depressurization Temp.	12.5°C
Blowdown line Depressurization Temp.	-13.3°C
Design Flow Rate	386 MMscfd

6.6. Equipment Description

6.6.1. Pig Launcher / Receiver

The first segment of the pipeline system; 24" x 6.7km Multiphase Pipeline from BBOU manifold to ELEPA manifold shall be fitted with a Pig Launcher and a Pig Receiver to facilitate its cleaning, inspection, and maintenance. The Pig Launcher shall be located at the BBOU manifold from where the pipeline starts while the Pig Receiver will be at the ELEPA manifold.

Also, the 36" x 31.5km segment of Multiphase Pipeline from ELEPA manifold to AKABEL shall be fitted with a Pig Launcher and a Pig Receiver to facilitate its cleaning, inspection, and maintenance. The Pig Launcher shall be located at the ELEPA manifold while the Pig Receiver will be at the AKABEL Station.

The Pig Launcher / Pig Receiver shall be designed for use of intelligent pigs. Design code for the Pig Traps shall be ASME B31.8 while the End Closure shall be designed to ASME Section VIII Div. 1 and additional requirements of Shell DEP 31.40.10.13 – Gen. Technical Specification – Design of Pipeline Pig Trap Systems.

The quick opening closure shall be equipped with a mechanical interlocking system to prevent opening of the Pig Trap when pressurized. Pig signallers shall be installed on both sides of the pig launcher outlet valve to provide confirmation that the pig was successfully launched and passed through the pig trap valve. A Catch Tray shall be placed directly under the End Closure to prevent oil or debris contamination of the surrounding environment. Pressure Safety Valves shall be installed on the Pig Traps for overpressure protection.

Provision shall be made in the layout for Monorail Trolley for Pig handling operation at BOUBUWE BOU, ELEPA and AKABEL

6.6.2. Flare System

The vent from the Pig Launcher PSV, the Pig Launcher manual depressurisation line and the pipeline depressurisation line will be tied to a vent stack located around the manifold platform. A manual ball valve shall be used to enable the depressurization operation and the flow rate shall be controlled by a globe valve downstream of the ball valve. It is pertinent to state that the pipeline depressurization via a manual depressurisation line will be carried out at BBOU, ELEM & AKAB end.

6.6.3. Drain Pan/Pit and Drainage System.

A Drain Pit, whose volume shall not be less than two times the volume of the Pig Trap shall be provided for the Pig receivers. It serves to collect drain from the Pig Traps drain pan install beneath

Commented [AJ26]: Epic to confirm whether Venting to atmosphere is allowed as per the Local laws.

Commented [KY27]: The venting is not a continuous venting operation. It is only required during pigging operations or during an emergency response to fire

Commented [H28]: The law permits operational venting.

the Pig trap door via gravity. The Drain Pit shall be covered to avoid rain water ingress and shall have a Level Gauge mounted on it. Provision for venting any trapped gas and for evacuation of the drain shall be made.

After pigging operation, inventory is evacuated from the pig trap through the vent and drain outlets. During normal operation, the pig trap is isolated from the process by closing all isolation valves upstream the trap. (Kicker line valve, balance line valve, pig trap isolation valve, vent/relief and drain valves). Positive isolation is achieved by means of a double block and bleed valve arrangement. The Drain Header is sloped towards the Drain Pit to enable complete draining of the Drain Header.

The drain pit at shall be evacuated by trucking/Vessel via a quick coupling connection mounted on the drain pit

6.7. Isolation Philosophy

The positive isolation of the Pig Traps and any pipeline segment shall be achieved by means of a double block and bleed valve arrangement together with a spectacle blind.

6.8. Pipeline Manual Depressurisation

To carry out maintenance activities on the Multiphase Pipeline system, the Pipeline system needs to be depressurized. To achieve this, a dedicated manual depressurisation line shall be provided at the manifold Stations and at Akabel. Suffice it to say that manual depressurization of the entire pipeline shall be carried out at Akabel station only. The manual depressurization line shall incorporate a Ball valve and a globe for flow control.

7. Pipeline Design Basis

7.1. 24" Multiphase Pipeline Design Data

The pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L – X70
Installation Temperature	23°C
SMYS	450 Mpa
NPS	24"
Density	7,850 kg/m ³
Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Specified Minimum Yield Strength (SMYS)	450 MPa
Ultimate Tensile Strength	535 MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
OD / Wall Thickness	TBA
Design Factor	0.6
Corrosion Allowance	3 mm (based on general provision and also as confirmed from the hydrocorr analysis)
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	40mm (to be confirmed)
Installation Condition	Trenching
Cathodic Protection	Impressed current

Commented [AJ29]: What is basis for selection of corrosion allowance.

Commented [H30]: Confirmed from hydrocorr analysis

Parameter	Description
Design Life	30 years

7.2. 36" Multiphase Pipeline Design Data

The pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L – X70
Installation Temperature	23°C
SMYS	450 MPa
NPS	36"
Density	7,850 kg/m ³
Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Specified Minimum Yield Strength (SMYS)	450 MPa
Ultimate Tensile Strength	535 MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
OD / Wall Thickness	TBA
Design Factor	0.6
Corrosion Allowance	3 mm (based on general provision, and confirmed from hydrocorr analysis). However, actual is expected to be higher than that of BBOU owing to fluid properties.
ANSI Class	900#

Commented [AJ31]: What is basis of 3.0mm corrosion allowance

Commented [H32]: Hydrocorr analysis

Parameter	Description
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	40mm (to be confirmed)
Installation Condition	Trenching
Cathodic Protection	Impressed current
Design Life	30 years

7.3. Pipeline Wall Thickness Calculation

The wall thickness of the pipeline system shall be calculated using a design factor of 0.6 for the entire stretch of the Pipeline Route. The calculation shall be carried out as per ASME B31.8 using the latest survey information to ascertain the adequacy of the selected wall thickness for the pipeline.

The selected pipe shall have a diameter to wall thickness (D/t) ratio that is not greater than 96

7.4. Route Selection

The pipeline shall be routed along the new ROW as acquired and surveyed. However, if due to design constraints, some section of the acquired ROW will not offer economical and safe design of the pipeline system, an alternative detour shall be proposed for the approval of Employer. In general, the optimal route shall be selected to minimize any impact on the environment, with due consideration to terrain, natural features such as rivers, road, flood plains and natural reserves, and Host Community considerations. The route optimisation shall consider the followings:

- Rivers and ravines should be crossed at right angles and in a straight run of the river away from caving banks, if possible. A safe distance of at least 3m shall be maintained between new and existing crossing(s) to avoid damage to existing pipelines during construction activities.
- Rough terrain and wetland crossings should be minimized.
- Road crossings should be minimized.
- Areas with potential for communities to expand beyond the current limits should be detailed, while areas with a high potential for community encroachment should be avoided. However, where pipelines must traverse these areas design factors shall be applied as advised by the project **quantitative risk assessment**. The FEED outcome will be validated during detailed engineering
- Pipelines should not be routed parallel and adjacent to power lines where reasonable alternatives are available.

Commented [AJ33]: EPIC to furnish the details of QRA study for Employer's Review

Commented [KY34]: Noted

7.5. Geotechnical Properties

Soil Investigation surveys have been conducted and the results are contained in document No: 6064/PLRS/001-PRP issued by BRONE POSITIONING & SURVEY LIMITED on behalf of BFPCL. The FEED will be carried out in accordance with the data provided in above referenced report.

Commented [AJ35]: EPIC to provide load intensity of structures. The load intensity depends on the load per column supporting platform for which, number of column proposed for platform to be shared with Brone.

Commented [KY36]: Noted. But this will not be a content in the BoD document. It is one of the FEED outputs and can only be determined when all mechanical, piping, electrical and structural components has been considered.

7.6. Pipeline Crossings

The design of Foreign Service, track, minor road crossings shall generally be based upon open cut construction techniques for installation of the new pipeline systems while major roads shall be crossed by thrust boring. Pipelines shall be installed in accordance with the specific details drawings. All new pipelines at crossing shall be installed below existing pipelines.

Crossing of existing pipelines, cables, power lines and road, to be accomplished generally below the existing service, shall be at an angle of between 60° and 90°.

Pipeline wall thickness for all road crossings (major, minor road and track crossings) shall be designed in accordance with API RP 1102 as uncased crossings. Due to the inherent problems of water ingress into cased crossing that leads to galvanic corrosion and the tedious nature of filling of pipeline casing with filling materials such as Cements, Bentonites and Petroleum wax, the choice to use uncased crossing is advocated.

Commented [AJ37]: What is the basis of selecting the uncased crossing? would it not be better if Sleeves or Casing is provided.

Commented [KY38]: The Basis for selecting uncased crossing is given in the next sentence. It states "Due to the inherent problems of water ingress into cased crossing that leads to galvanic corrosion and the tedious nature of filling of pipeline casing with filling materials such as Cements, Bentonites and Petroleum wax, the choice to use uncased crossing is advocated"

The live, dead and impact loadings at all road crossings shall be calculated as outlined in API RP 1102 with tandem axle arrangement. Acceptable criteria for total effective stress in pipeline road crossing calculation shall be $S_{eff} \leq 0.72 \times SMYS$ in accordance with API 1102.

Crash barriers shall be installed at all road crossings to protect the installed pipeline(s) from third party damage in accordance with Project typical drawings.

7.6.1. Marsh, Swamp and Mangrove Swamp Area Crossings

In marsh and swamp areas, the design of the pipeline system shall follow the requirements and regulations of the Inland Waterway Authority and other governing bodies.

In marsh and swamp areas the pipeline shall be buried in the trench. In areas where there is risk of erosion due to slopes or high currents, replanting of vegetation, installation of current breakers etc, shall be considered. The methodology for pipeline construction and installation shall be detailed in the document; Pipeline Construction and HDD Installation Specification - EA-BFPCL-GPMC-GGPLFEED-PPL-SPC-009.

Commented [AJ39]: In Swamp area, the soil may get eroded during course of time and thereby soil cover will be reduced. What is the EPIC's experience for such eventualities.

Commented [KY40]: You are right and which is why EPIC stated in the next sentence that current breakers will be provided in areas where there is risk of erosion.

Various techniques may be adopted depending on the local geography. These fall into three broad categories:

7.6.1.1. Dewatering:

Should the need arise for swamp land to be dewatered for the purpose of trenching, water shall

be pumped from underground wells to lower the water table on a local basis thus minimising any environmental impact.

7.6.1.2. Building a Causeway:

Where pipe laying is to be carried out to one side or within a causeway, the former is preferable from an environmental standpoint as the causeway, being a permanent feature can be narrower.

7.6.1.3. Methods Involving a Dredged Ditch:

The pipeline is assembled on a land based or floating platform and progressively pulled and pushed into a channel already dredged. Alternatively, a whole section may be pre-assembled, floated out and sunk in position.

The long-term impact would depend on whether the resulting ditch is backfilled or left as a waterway.

7.6.2. Waterways Crossing

At rivers, creeks, canals, other waterways crossing the pipeline will be buried in the natural bed of the waterway.

The minimum cover of the pipeline shall be as follows:

7.6.2.1. Watercourses of Width Less Than 60m:

The minimum cover will be 2.0m from the lowest bed level for the full width of the crossing.

7.6.2.2. Watercourses of Width Larger Than 60m:

The minimum cover of pipeline shall be 3.0m below LLWS or 2.0m below the lowest bed level whichever results in the greater depth or as indicated on construction drawings.

For the watercourses of width larger than 60m a design factor 0.6 SMYS shall be adopted throughout the crossing.

7.6.2.3. Navigable Rivers and Marine Waterways:

The minimum cover of pipeline shall be 4.5m below LLWS or 3.5m below the lowest bed level, whichever results in the greater depth or as indicated on construction drawings.

For the navigable rivers and marine waterways a design factor 0.6 SMYS shall be adopted throughout the crossing.

7.7. Pipeline Depth of Cover

The pipeline shall be designed for the following minimum cover in table 5: DEP (31.40.00.10-Gen)

Table 5.4: Pipeline Depth of Cover

Commented [AJ41]: EPIC is requested to share this DEP

Commented [KY42]: The DEP is a set of SPDC document. The right to share and circulate rests with SPDC. TCE can request for the DPEs through BFCL from SPDC for access to the DEP databank.

Location	Minimum Depth of Cover
Normal Terrain	1.0 m
Swamp and Mangroves Areas	1.0 m
Industrial & Populated Area	1.2 m
City/Town Area	1.2 m
Asphalt Road Crossing (metaled)	1.5 m
Railway Track	1.2 m
Agricultural Area	1.0 m

7.8. Pipeline Separation / Safety Distance

In the case where the new pipeline is laid alongside an existing pipeline on the same ROW, the lateral distance separating the new and existing pipeline shall not be less than 15m unless otherwise specified by QRA. This shall be further evaluated during detailed engineering when QRA shall be carried out.

The lateral spacing between two new lines running parallel to each other shall be as follows:

- Dry land 3 metres minimum
- Swamp land 10 metres minimum

7.9. Pipeline Bend Requirement

Where possible, all major changes in direction shall be achieved by cold bending the linepipe in the field. The minimum bend radius for field bending shall be 40 times the nominal diameter of the pipeline (40D).

Factory manufactured hot induction bends may be permitted where there is a limitation of space for using cold field bends. Factory made induction bends are permitted up to minimum bend radius of 3 times nominal pipe diameter (3D) and is adequate for PIG movement.

Elastic bending shall be utilized for gradual changes in elevation achieved by profiling the trench and laying the pipeline to fit the contour without cold bending. The minimum elastic bend radius shall be calculated based on the maximum permissible combined stress. Bending stresses due to elastic bending shall be considered to be additive to maximum thermal longitudinal stresses.

No tie-ins shall be made within any section of bend. Metered bends shall not be used.

7.10. Pipeline Stress Analysis

Preliminary Stress Analysis shall be carried out for the pipeline system in accordance with ASME

Commented [AJ43]: This clause is not clear, kindly further elaborate

Commented [H44]: See the spacing as provided in the bullet points

Commented [AJ45]: Please ensure that 3D Bend is adequate for pig movement.

Commented [H46]: Noted

B 31.8. Pipeline shall be checked for structural integrity considering the anticipated stress conditions in the pipeline during installation, hydrostatic testing, and operation.

Secondary stresses such as traffic load, longitudinal stress, hoop stress etc. for buried pipeline shall be checked in accordance ASME B31.8 to ensure that the combined stresses do not exceed 90% of SMYS of pipe material.

The buried pipeline analysis shall use soil friction, soil stiffness and backfill density derived from the geotechnical survey. The model shall accurately reflect the geometry of the pipeline system.

Pipeline Stress Analysis carried out by computer software program shall use CAESER

Expansion of buried pipelines due to pressure and temperature effects shall be restrained to acceptable limits. The restraint shall be provided by friction between soil and pipe and/or by the use of anchors of the full thrust or drag type. At the transition between below-ground and above-ground, the expansion shall be limited by consideration of the pipeline approach or by utilization of a pipeline anchor block. Preference shall be given to designing the pipeline approach to limit aboveground movement over the use of anchor blocks.

Expansion shall be considered acceptable when it does not overstress end or branch connection pipeline or cause any external soil disturbance or pipeline coating damage.

7.11. Upheaval Buckling Check

A pipeline subject to high internal pressure and temperature will tend to expand. Since the pipeline is buried, it is not free to expand; thus, resulting in axial compressive force. If this force exceeds the vertical restraint against uplift movement created by the pipe's weight, bending stiffness and resistance by soil cover, the pipe will tend to move upwards.

The problem of upheaval buckling of pipelines occurs for onshore pipelines of oil and gas production fields. High pressures and temperatures increase the danger of upheaval buckling. Also, pipelines going through areas of overbend increase the risk of upheaval buckling.

Upheaval Buckling Calculation of Pipelines shall give the recommendations for assessing the risk of upheaval buckling of pipelines and provides alternatives for corrective action prior to or either after pipeline construction to prevent upheaval buckling. Berm height shall not be considered for pipeline upheaval buckling calculations. Upheaval buckling of onshore pipeline can be mitigated by any or a combination of the following methods:

- Profile Smoothing
- Increased flexibility
- Changes in operating parameters
- Reduction in wall thickness

7.12. Pipeline Stability

Anti-buoyancy measures shall be taken by providing external concrete coating over the corrosion coating in the event that the pipeline is laid in high water table or marshy/swampy areas.

Special considerations shall be given to pipelines installed in weak soils, culvert entries, etc., where differential settlement may lead high stress levels and possible loss of pipeline integrity by providing pipeline anchors and expansion loops.

The resulting negative buoyancy shall be sufficient to prevent the combined action of hydrostatic forces.

The pipes concrete weight coating shall be of a thickness ensuring a minimum negative buoyancy of 1.2.

7.13. Pipeline Valves

Pipeline Valves such as pig trap valves shall be designed in accordance with requirements of API 6D.

7.14. Isolating Joints

Aboveground portions of pipeline which has buried sections with impressed current cathodic protection shall be electrically isolated using monolithic isolation joint. All isolation joints will be installed above ground, immediately after the pipeline buried/ above ground transition. Isolation Joint shall be designed in accordance with requirement of Technical Specification for Pipeline Isolating Joints.

7.15. Pipeline Flanges, Fittings and Branch Connections

Material for pipeline flanges and fittings shall be compatible or equivalent with mainline line pipe material. Flanges, forged fittings, gaskets, bolts & nuts, shall comply with applicable codes and standards and project specifications.

Slip-on flanges, mitre connections and threaded connections (pipe to pipe fittings) shall not be used in any part of the pipeline system.

Pup pieces for any part of pipeline shall not be less than 1000 mm.

For mechanical strength, there shall be no branch or instrument connection smaller than 2" on any section of the pipeline.

7.16. Pipeline Construction Requirement

7.16.1. General Pipeline Construction Requirement

In general, pipeline construction shall be carried out in accordance with Technical Specification

for Pipeline Construction.

7.16.2. Right of Way

Site clearance shall involve stripping of all surface and subsurface vegetation, destumping and removal top soil. The use of a layer of compacted sand-cement is recommended to facilitate trafficability during construction and for temporary drainage of surface water run-off. All these activities shall be in accordance with agreed standards

Effective management of the ROW with respect to spacing shall be considered. The width of construction ROW and permanent ROW shall be indicated in the project drawings.

In areas where the soil is so loose or unstable to be used for backfilling, the use of foreign soil borrowed from elsewhere where the soil is more compact shall be considered for backfilling.

ROW shall be reinstated with the stripped and preserved top soil to restore the surface as close as possible to the original condition before construction works.

Commented [AJ47]: EPIC has reviewed the Geo Technical Report and EPIC to confirm the soil is good for back filling

Commented [H48]: Material to be evaluated for suitability before use

7.16.3. Pipeline Welding

Pipeline welding shall be carried out in accordance with API 1104 technical requirements and Technical Specification for Pipeline Welding". All pipeline welds shall be 100% radiographed. All pipeline tie-in welds shall be 100% ultrasonically tested in addition to 100% radiography.

7.16.4. Pipeline Warning Sign / Permanent Markers

Pipeline warning signs shall be installed in accordance with EMPLOYER and Project requirements. Generally, pipeline warning signboards shall be provided at the following locations:

- Major horizontal direction changes in pipeline route alignment
- Each side of major and asphalt road crossings
- Crossing of existing utilities
- 0.5 km intervals along the pipeline route through remote terrain
- 250 m intervals along the pipeline route near fenced areas and through populated areas

Warning, direction and danger signboards, as applicable, shall be installed.

In order to enhance pipeline safety against third party accidents a plastic warning tape shall be installed all along the pipelines bearing the description: "High Pressure Gas Pipeline " as applicable. The warning tape shall be installed 300 mm above the pipeline top before completing the backfilling operation. The tape shall be yellow / orange color with black letters and the description of the tape shall be in English.

7.16.5. Pipeline Hydrostatic Test and Pre-commissioning and Commissioning

Hydrostatic testing of the pipeline shall be carried out after construction to prove their strength and tightness at field. Hydrostatic testing and commissioning shall be carried out in accordance with Specification for Pipeline Hydrostatic Test and Pre-commissioning and Commissioning

8. Piping Design Basis

8.1. General

Piping systems shall be designed, fabricated, and tested in accordance with ASME B31.3, and additional requirement of **Shell DEP 31.38.01.11-Gen_- Manual - Piping - General Requirements**, for all piping at the manifold stations and Akabel.

Screw fittings shall be avoided for high pressure hydrocarbon systems.

Piping design shall be carried out with due consideration given to safety, accessibility, constructability, operability, maintainability, reliability, and life-cycle cost.

Threaded connections are not allowed in hydrocarbon systems. Threaded connections can only be used in hydrocarbon processing areas on instrument impulse lines after the DB&B isolation valves. The DB&B will be flanged on the process side, and threaded on the instrument side.

Use of small bore piping ($< 1 \frac{1}{2}$ ") in process systems shall be minimized due to its susceptibility to mechanical damage.

Piping fittings, etc., should not overhang the skid limits as these will be susceptible to damage during transit.

The number of flange connections shall be minimized in order to reduce potential emissions and for safety reasons.

8.2. Design Conditions

If a system normally working/operating above 0°C is connected to a system operating at 0°C or below, the first 50 OD (with a maximum of 10m) shall be designed to the specifications of the lower temperature system.

8.3. Piping Design Engineering Principles

8.3.1. Piping Wall Thickness

- All piping component wall thicknesses to be specified shall be calculated in accordance with either ASME B31.8 or ASME B31.3. Pipe wall thickness is based on the design conditions specified in the various process data and the P&IDs.
- The calculation of wall thickness shall account for any additional loadings imposed, such as thermal loads due to thermal expansion, sustained load due to weight, live loads, and other loadings that must be considered per ASME B31.3/ASME B31.8. The piping stress engineer shall account for the imposed loads that shall be considered per the design code.
- Nominal sizes and wall thicknesses of pipe, shall be in accordance with ASME B36.10M and ASME B36.19M, as applicable.

Commented [AJ49]: EPIC to share this Document

Commented [H50]: TCE to request access to document through BFPCL

- d. The basis of design of butt-welded fittings not covered by ASME B16.9 shall be determined during detailed design phase.

8.3.2. Pressure/Temperature Ratings

- a. Pressure/temperature ratings of flanged components in sizes NPS 3/4 through NPS 24 shall be based on ASME B16.5.
- b. Under normal conditions, flanged components larger than NPS 24 shall be specified as conforming to ASME B16.47 Series A for Class 300 and higher.
- c. Pressure shall be indicated in psig (kPag or barg) with temperature in degrees Fahrenheit (Celsius), unless otherwise specified in the piping deliverables as applicable.
- d. The design limits for all mechanical/piping equipment and components conform with the pressure/temperature ratings capability as defined in the applicable equipment/component standard or equipment/component material description.

8.3.3. Design Values

- a. Allowable design stresses for piping components shall be as specified per the allowable stress tables in ASME B31.3 as applicable. The use of other values supplied or accepted by the owner is permitted.
- b. Internal pressure calculations for seamless pipe as well as for 100 percent radiographic tested (RT) welded pipe shall be performed based on a weld joint factor $E=1.00$ and a wall thickness mill tolerance of 12.5 percent.

8.4. Facility Layout Philosophy

The plant layout shall comply with the main following design criteria.

8.4.1. Plot Plan

The plot plan (facility layout) gives detailed information on the facility, in order to facilitate:

- a. the definition of the topographic position of all equipment and the classification of hazardous areas
- b. a check on the interference among equipment, structures and civil works, with a definition of those equipment located in buildings
- c. a control on the easiness of erection, on the lifting equipment availability, clearances and maintenance zones all around the equipment
- d. a detailed definition of the plant planimetric drawings
- e. a definition of main structures of the plant in conjunction with the main equipment position and of the related weights
- f. a preliminary control of the suitability of the main layout based on safety criteria
- g. a preliminary control of the escape routes
- h. a detail definition of all the other plan drawings of the facility to be designed

- i. a definition of the foundation drawings

The plot plan is also used for the production of a drawing indicating the plan division into areas. Such drawing is called "Key Plan".

8.5. Piping Material Specification and Classification.

Piping systems design and selection of materials shall accommodate all pressures, temperatures, stresses, dynamic forces, corrosive conditions, etc. that could occur both during fabrication, transportation, erection as well as a result of extreme conditions experienced during startup, normal operations, shutdowns, and emergency situations.

A piping material classification shall be done for grouping of piping items according to pressure and temperature limits and similar process characteristics in accordance with **DEP 31.38.01.10-Gen** and **DEP 31.38.01.84-Gen**.

A new classification shall be done for all the in-plot pipeline components installed after the pig traps using an adequate DEP classification as the basis.

8.5.1. Material selection for piping

Material: Carbon steel and Stainless steel (Note-1)

Corrosion allowance: **6mm for carbon steel** (Hydrocorr analysis) and 0mm for stainless steel

Note-1:

The grades of steel required for the various piping requirements shall be as specified in the project Piping Materials Classes defined in the specific specification as required according to the Process Design Basis.

8.5.2. Change of Piping Material Classes

When a pipe of a given class is connected to a pipe of a higher class, the portion of line extending from the connection up to and including the first valve installed shall be made in the higher class.

8.6. Pipe Sizes

The following pipe sizes shall be used as may be required:

DN 15	½ "	DN 100	4"	DN 400	16"
DN 20	¾ "	DN 150	6"	DN 450	18"
DN25	1 "	DN 200	8"	DN 500	20"
DN 40	1 ½ "	DN 250	10"	DN 600	24"
DN 50	2"	DN 300	12"	DN 750	30"
DN 80	3"	DN 350	14"		

Commented [AJ51]: Please provide this document for reference

Commented [KY52]: The DEP is a set of SPDC document. The right to share and circulate rests with SPDC. TCE can request for the DPEs through BFCL from SPDC for access to the DEP databank.

Commented [AJ53]: Basis for 6mm? why the selected corrosion allowance is different for pipeline and piping

Commented [H54]: Hydrocorr analysis

8.7. Piping Components

Piping components, fittings, valves, gaskets and bolting shall comply with the project technical specifications of the component. The use of special materials and/or piping components (otherwise called Special Piping Items) not included in the piping classes shall be minimized, and shall require the approval of the responsible piping material engineer.

If a required item is not covered by the piping classes and requires individual calculation or selection, an SP number shall be allocated and used on all relevant documents, including the P&IDs. Such items might include valve locks, non-standard valves, hoses, transition pieces, quick release coupling, etc.

8.7.1. Line size

Pipes and fittings having sizes of 1 ¼", 2 ½", 3 ½", 4 ½" and 5" will not normally be used. Where such diameters are required for connections to equipment, the pipe length will be reduced to a minimum.

8.7.2. Flanges

The use of flanges on pipes will be minimized and anyway limited to:

- connecting lines to equipment
- Connecting flanged elements installed on line (valves, etc.)
- On pipes which have to be removed for particular service and maintenance

Flat faced flanges (FF) will be used for cast iron valves and/or equipment nozzles joints

8.7.3. Elbows

Changes in direction will be made by means of elbows, forged or bent pipes. The elbows shall be long radius type unless otherwise specified.

8.7.4. Other piping components

The following rules must be stated taking into account installation, maintenance, operability in normal and emergency service and upset conditions

a. Valves:

Requirements will be provided for the following type: gate valves, globe valves, check valves, plug valves, ball valves and butterfly valves.

b. Vents and drains:

Piping shall be designed in order to avoid gas and/or liquid accumulations.

Otherwise:

- i. On all liquid accumulations, drain valves will be installed
- ii. On all gas accumulations, valve vents will be installed

8.7.5. Control valves

In control valve installation, sufficient clearance for actuator accessibility and/or removal shall be foreseen. Control valves should be positioned preferably with the stem oriented vertically. The stem should not be oriented horizontally or downward, particularly, layout should be analyzed case by case.

8.7.6. Safety valves

Safety valves shall be installed in accessible areas in order to ensure operability and routine disassembling. Safety valves shall always be installed vertically.

Valves discharging to atmosphere will be located at the highest point of the line on which they operate.

Valves discharging into a closed system will be located at a minimum distance from the header to which they are connected.

8.7.7. Gaskets

- The correct gasket for the flange class and process conditions shall be selected in accordance with relevant specifications and piping material requirements.
- Gaskets for raised face (RF) flanges shall center using an outer guide ring. RF gaskets shall be used for class 600# and below
- Spiral wound (SW) gaskets shall be specified to have inner and outer rings.
- Octagonal Ring Joint gaskets shall be the type with an octagonal cross-section

8.8. Layout Requirements

Piping layouts shall be carried out with the intention of limiting the number of field welds required whilst at the same time maximizing shop spool fabrication.

Piping layouts shall be sufficiently flexible and accessible enabling piping or mechanical equipment to be removed without imposing any undue stress on adjacent piping or equipment.

All piping shall be arranged to provide the specified headroom and clearances around equipment for ease of access during operation, inspection, repair, and maintenance.

Particular attention shall be paid to clearances required for the removal of equipment without the need for disturbing adjacent piping.

Piping shall be kept clear of all openings, inspection points, overhead cranes, runway beams, davits, laydown areas, space allocated for instrument removal, normal access ways, and emergency egress routes.

Piping design shall ensure that all valves and instruments are accessible by safe means for operation and maintenance/repair/removal purposes through use of suitable access platforms,

removable platforms & ladders, etc. Human factor engineering shall be incorporated in the design. PDMS 3D model shall be developed for the associated piping inclusive of the pig traps at the manifold station. Based on DEP 30.10.05.11_Gen- Plant Model Construction and Review and approved 3D Interface procedure.

8.9. Supports

Pipe supports design shall be based on the maximum loads to which the support is subjected, considering all conditions analyzed.

Commented [AJ55]: EA is requested to share this standard for reference

Commented [KY56]: The DEP is a set of SPDC document. The right to share and circulate rests with SPDC. TCE can request for the DPEs through BFCL from SPDC for access to the DEP databank.

9. Material Specification and Corrosion Management

Generally, materials selection for all components will conform to applicable Industry and Shell Standards. The underlying principle to be adopted for materials selection and corrosion control, optimization of material and equipment life cycle costs shall be defined in this project FEED documentation.

Brittle fracture (Failure Mode and Effect Analysis) (FMEA) shall be managed in the facility through the application of Minimum Design Metal Temperature (MDMT) (Minimum Design parameters) limits relative to the Minimum Metal Temperature (MMT) (Minimum factors essential to maintain the facility integrity).

External corrosion is managed through a combination of cathodic protection, coatings and appropriate material selection. (External corrosion management is subdivided into three zones: buried and submerged systems corrosion, tidal/splash zones corrosion, and atmospheric corrosion).

The materials selection and specification shall be made to ensure the fitness-for-purpose of each element, in respect of threats to the integrity and functionality of the facilities.

9.1. Specifications and Corrosion Management & Controls

In the bid to achieve the desired degree of protection at optimal cost, the following factors shall be considered:

- Design life
- Properties of internal fluid
- Surrounding environment
- Mechanical properties
- Wall thickness
- Weldability
- Fabrication and installation methods and procedures
- Market availability of materials
- Economics

The materials listed below have been considered from which Mild Steel (MS) has been selected, but with provision for inhibition:

- Mild Steel (MS)/Carbon Manganese Steel
- Various CRAs (Austenitic SS, Ferritic SS, Martensitic SS, Duplex SS)
- Precipitated Hardening Steel (with Al)
- Pipeline with internal CRA lining or cladding
- Composite materials and non-ferrous materials

Specifically BBOU to AKAB requirement will be as follows:

S/N	Description	Design Information	Feed Specification	Corrosion Management & Control	Remarks
1	Design life	30 Years	All specs to be dev during FEED	Cradle to grave mgt.	
2	Internal fluid	Wet gas	Confirm conceptual rates	Corr. analysis	
2.1	Materials selection	MS Plus Inhibition			
2.1.1	MS – BUBUOWE to ELEP	CA – 3mm + inhibition	16" API 5L X70	Inhibition injection and monitoring	
2.1.2	ELEP to AKABEL	CA 10mm + inhibition	24" API 5L X70	Inhibition injection and monitoring	
2.1.3	Inhibitor	Nelco/Champion X	Choice from NKK033, CK377 and EC1304	Inhibition injection and monitoring and testing	
3	External surrounding	Extremely corrosive	Spec to be established after design	coating design and CP design	
3.1	Chemical test	Chlorine/chl oride presence			
3.2	Coating specs	Confirm concept	Epoxy/PE/PP (Spec code NACE 06053 / ISO 21809-3)	Isolation and protection from environment	
3.3	CP design	Soil resistivity data	Establish specs for C.P G.B crossings (river & roads) temporary anodes and equipment.	Manage design construction and quality and establish requisition	

9.2. External Surface Coating & Painting (Buried & Submerged Facilities Protection)

The buried and submerged facilities are managed through a combination of cathodic protection and coating (3LPE) of the selected materials.

9.2.1. Cathodic Protection System

An effective Cathodic Protection System shall be provided to prevent all buried steel structures, and internal surfaces where deemed applicable. Considerations for this system will cover types of protection, capacity, and the equipment specification for the location.

Cathodic protection system shall be installed to protect buried pipelines and bulk lines from excessive corrosion throughout the life of the facilities. The design shall be based on a site survey to obtain information on the terrain, soil resistivity, stray currents and other possible interactions with neighbouring pipelines' cathodic protection systems.

Sacrificial or impressed current cathodic protection or combination of the two H(Hybrid systems) shall be used. A suitable anode ground bed (s) shall be constructed. Test points shall be provided along the pipelines at an average of one or less kilometres, subject to accessibility and design of Cathodic Protection System.

Isolating joint shall be installed at specific location to ensure electrical discontinuity of the buried pipelines from the above ground facilities.

9.2.2. Coating of Submerged and Facilities:

Coating of bulk lines, piping, pipelines and other equipment that are buried and submerged shall be in accordance with the relevant Shell and international standards and specifications.

9.3. Protection of Equipment at Splashzone Area:

Materials within this zone shall be protected with special splash zone coating that can withstand wear, tear, abrasion etc. (SplashTRON)

9.4. Painting of Equipment in Atmospheric Area:

The painting and coating of, piping, equipment and other utilities for the proposed facilities shall be in accordance with the relevant Shell and international Standards and Specifications.

The external corrosion protection for the above ground end facilities such as piping, manifold, launchers, receivers, etc. shall be provided for by painting system as specified in the specification for painting and coating in consultation with the manufacturer and approved by BFPCL

For the (painting) of insulated piping and equipment, (rich) zinc-based (painting systems) may be

implemented to provide protection against corrosion (including) under insulation (CUI).

9.5. Pipeline/Flowline Internal Corrosion Control

Risks of internal corrosion in the multiphase pipeline shall be essentially mitigated by regular pigging, periodic / continuous injection of corrosion inhibitors (depending on MSDS), and provision of adequate corrosion allowance in pipe wall thickness and by selecting the right materials.

9.6. Remote Corrosion Monitoring Sensors

Remote corrosion sensors capable of relaying spot UT measurements in real time to remote locations shall be deployed as part of the corrosion management strategy for the pipeline system. Installation and mapping schematics for the sensors shall be developed as part of FEED which shall be fully developed in detailed engineering. Other real time corrosion monitoring tools may also be evaluated as part of FEED design if necessary.

Lessons learnt from the deployment of the remote sensors at other Employers locations will be factored into the design of the sensors.

Commented [AJ57]: Please clarify whether the Corrosion Inhibitor will be injected continuously OR periodically?

Commented [KY58]: This is a provision that can only be provided at the DED stage when more clear data are available. The injection skid OEM's input on critical design considerations are needed to make a judgment such as MSDS, optimal dosing rate etc. So at this FEED stage and for BoD, the flexibility is provided for periodic or continuous.

10. Instrumentation Basis of Design

10.1. General

The pigging operation of the pipeline will be manned, controlled and monitored locally. Pig passage will be detected locally using non-intrusive pig detectors. On-line pressure and temperature gauges shall be used to monitor the pipeline pressure and temperature respectively.

Overpressure protection for the pig traps and bypass line at the stations shall be by pressure relief valves.

The stations shall have safeguards meeting the requirements of API RP 14C, for overpressure protection of the pipeline.

Overpressure protection of any downstream facilities tied to the pipeline at shall be done by a Safety Instrumented System (SIS) and a Pressure Relief Valve (PRV) according to the requirement of API RP 14C. The SIS shall be the primary overpressure protection while the PRV shall be the secondary overpressure protection.

Depressurisation of the pipeline shall be manual, planned and done at ELEP and AKABEL ends of the pipeline system.

The Process Automation and Control (PACO)/ICSS system of the Station shall be tied to the PACO/ICSS system of the Plant at AKABEL. All required hardware and software modification for the integration shall be done.

The Fire & Gas Detection System (FGS) of the Stations shall be tied to the Fire & Gas Detection System of the Plant at AKABEL.

10.2. Pressure Measurement

Pressure measurement at the pig trap stations shall be through dial type pressure gauges. The pressure gauges design and construction shall conform to ASME B40.100 Grade 2A, and accuracy of pressure gauge shall be $\pm 1\%$ Full Scale. The maximum process operating pressure shall be 50% - 75% of the pressure gauge full scale range.

Pressure gauges shall be liquid-filled or dry type with integral pulsation dampener. Sensing element of pressure gauges shall be stainless steel bourdon tube, and other wetted part materials shall also be stainless steel.

Pressure gauges shall be supplied together with stainless steel pressure snubber and double-block-and-bleed or block-and-bleed RF monoflange, as process isolation valve.

10.3. Temperature Measurement

Dial type, adjustable angle, bimetallic thermometers will be used for temperature monitoring at

both pig trap stations. Design and construction of the thermometers shall be as per ASME B40.3 and the normal operating process temperature shall be within 30%-70% of the upper range value of the thermometer.

The thermometers shall be installed through bar stock tapered thermowell and shall be removable in service. Thermowell material shall be stainless steel, its design and geometry shall meet the requirements of Part II, Section 3.9 of DEP 31.38.01.24-Gen. At all process conditions, the ratio of the Strouhal frequency and the thermowell natural frequency shall be less than 0.8, as stated in ASME PTC 19.3.

10.4. Level Measurement

Top mounted magnetic level gauge c/w bicoloured rollers will be used for monitoring hydrocarbon liquid level in pig trap drain pits. The level gauges chamber shall be furnished with vent plug for venting purpose. Body material of the level gauges chamber shall be stainless steel but process connection shall be carbon steel, to prevent galvanic corrosion. Both chamber and process connection material of the level gauges could be made of carbon steel.

10.5. Overpressure Protection

Overpressure protection for the pig traps and bypass line at the stations shall be by pressure relief valves. The pressure relief valves could be conventional spring type, balance bellows types or pilot operated types. The design and construction of the pressure relief valve shall meet the requirements of DEP 80.36.00.30-Gen, API 520, API 521 and API 526.

The Alarm layer shall comprise an analogue 4 – 20 mA, 24 Vdc, Ex ia pressure transmitter with self-diagnostic capability. The purpose of this transmitter is to notify operations personnel via HMI alarm display of attainment of maximum allowable operating pressure in the pipeline. This transmitter shall be connected to the Process Control System at BFPCL plant at AKABEL.

The Primary Protection layer at the stations shall meet the requirements for overpressure protection of Departing Pipeline stated in API RP 14C. The SIS shall comprise one analogue 4 – 20 mA, 24 Vdc, Ex ia pressure transmitter with self-diagnostic capability, and a tight shutoff, full bore, API 6D ball valve c/w pneumatic, single-acting, spring-release, scotch-and-yoke actuator assembly (Operation Shutdown Valve, OSD Valve). The pressure transmitter: OSD Valve's solenoid valve and limit switches (integral to the actuator assembly) shall be linked to AKABEL SIS panel. If the pressure transmitter records pressure of Set Point: Pressure High High in the SIS panel, the SIS panel shall de-energise the actuator's solenoid valve to shutdown (close) the OSD Valve, thus isolating (protecting) the pipeline from the overpressure.

The OSD Valve shall be designed and constructed to fail close (FC) when there is loss of instrument air supply, loss of electrical power or mechanical fault within the actuator assembly. Operations personnel shall be able to monitor and control the OSD Valve from the CCR in AKABEL.

Secondary Protection layer is the last resort when the Primary Protection layer fails. It shall comprise a PRV installed upstream the tie-in points of the pipelines at AKABEL, as recommended by API RP 14C. The set point of this PRV shall be determined as the FEED progresses and it shall vent pressure in excess of PRV set point to ABEL venting or flare system to ensure the downstream facilities do not encounter pressure in excess of the set pressure.

10.6. Pipeline Depressurisation

Depressurisation of the pipeline shall be manual and planned. It shall be done at the depressurisation platform of ELEM manifold Station node of the pipeline or at AKAB using a hand wheel operated globe valve to control the depressurisation.

10.7. Pig Detection

PIG passage during pigging shall be detected using battery-powered non-intrusive PIG detectors. The PIG detector shall comprise an electronic notification module and a magnetic and/or ultrasonic transducer (sensor). The electronic module should be furnished with Liquid Crystal Display (LCD), LED indicators and piezoelectric sounders (if required).

Transducers shall be strapped on (mounted on) the pipeline and the pig traps with the aid of their mounting brackets. The notification electronic modules can be strapped on the pipeline, mounted on 2" OD pipe supports or handheld. Electronic notification module that is meant to be strapped on the pipeline or mounted on 2" OD pipe support, shall be supplied with its mounting bracket.

The cable and cabling between the transducer and the notification electronic module shall be vendor (OEM) standard. Wireless communication technology could be used for connecting the transducer and the notification electronic module.

The PIG detector shall be suitable for use in zone 2 area and rated explosion proof (Ex d), as per IEC 60079.

10.8. Cables and Cabling

Cable core sizes shall be 1.5 mm², pair colours shall be red and black, outer sheath colour shall be black (or grey colour) for solenoids and limit switches cables, but blue for all intrinsically safe 4 – 20 mA, 24 Vdc analogue instruments cables.

Cable conductor shall be stranded plain annealed copper wire, insulation material shall be XLPE, shielding shall be overall by MS c/w drain wire, armouring shall be SWA and outer sheath material shall be PVC. Rated maximum operating voltage of cable shall be 300 V. Cables shall meet the requirements of EN 50288-7 and DEP 32.37.20.10-Gen.

Screens of instrument cables shall be insulated at the field instrument side and shall be connected to an instrument earth terminal adjacent to the signal pair. Screens shall be earthed only at the

marshalling cabinet (control panel) in the NLNG CCR.

Cables shall be supported and protected against mechanical damage by cable trays and cable trays shall not be supported by the pig traps, its associated piping or the pipeline.

Stainless steel cable glands shall be used for cables entering the field instruments and junction boxes. All cable glands shall be barrier type and certified Ex d.

Cables shall be furnished with labels and cable markers in accordance to DEP.30.00.60.21-Gen.

10.9. Fire and Gas Detection System (FGS)

The FGS of the pipeline Stations shall comprise UV/IR Flame Detectors, IR Point Gas Detectors and Open Path Gas Detectors. The FGS shall be integrated to AKABEL FGS; and all applicable FGS detectors shall be compatible with that of AKABEL Fire Alarm Control Panel.

The perimeter of the PIG Receiver Station shall be monitored with Open Path Gas Detectors, and flanged joints of possible gas leakage shall be monitored with IR Point Gas Detectors.

At least three UV/IR Flame Detectors shall be mounted in the PIG Trap Stations to ensure the stations are adequately covered in flame detectors cone of visions and to ensure a 2oo3 voting logic of the flame detectors, to avoid spurious alarm and/or trip signal.

10.10. Emergency Shutdown

The pipeline shall use a separate tight shutoff, full bore, API 6D ball valve, c/w pneumatic, single-acting, spring-released, scotch-and-yoke actuator assembly (ESD Valve), for emergency shutdown when fire is detected in the Stations.

The shutdown philosophy of the Stations shall be integrated into AKABEL emergency shutdown philosophy.

11. Telecommunications Design Basis

11.1. Basic Requirements

The telecommunication system shall provide reliable communication that are relevant for the pipeline system monitoring, operation, management, administration and maintenance, within the stations, between the stations, and between the stations and Central Control Rooms located at AKABEL and SPDC office in Port-Harcourt. The networks for process data communication shall be entirely separated from voice, security and office networks, and the communication networks for inhabitants of the residential areas (if any), having only a minimum number of secured interfaces, if necessary.

The following telecommunication services shall be provided at and in between all stations and the Central Control Rooms:

- Civils and Electrical site preparations for Redundant process data communication (SCADA)
- Fiber Optic cable system.
- **Structured** Office Cabling system for VOIP telephone and Data; to be transmitted to main control room in Akabel through fibre optic cable with VSAT system as back-up.
- Access control systems at the pipeline stations.
- Intruder Detection System at the pipeline stations.
- CCTV systems at TGS pipeline stations.
- Tie-in to the **office** LAN System at AKABEL being provided by OTHERS.

The following pipeline monitoring services shall be provided along the pipeline:

- Civils and Electrical site preparations for 3rd party intrusion detection in pipeline stations.

11.2. Fiber Optic Cable

96-Core (to be confirmed) micro fiber Single Mode Fiber Optic Cable shall be used for telecommunication purposes and shall also be the backbone of Data Transmission System (DTS). Fiber(s) of the same FOC shall be used for leak detection, pig tracking, and intrusion detection by the Fiber Optic-based Pipeline Monitoring System. 36 cores shall be for the project while 60 cores shall be reserved party BFPCL and SPDC and other third party. The different sets of fibers shall be demarcated and terminated in different dedicated ODFs at all the pipeline stations.

Fiber Optic cable shall be installed along the pipeline route from BBOU manifold Station, through ELEPA manifold stations to AKABEL Station. The number of Manholes to be installed along the ROW shall be determined in the course of this FEED

The FOC shall be installed in the same trench as the pipeline (blown in HDPE protective conduit). Typical drawings shall be produced to show the position and mounting of the FOC relative to the pipeline.

Commented [LMS59]: Whether service providers have towers near the manifolds for enabling telecommunication from gas manifolds to Brass Island?

Commented [H60]: FoC with radio frequency tower as back up

Commented [EL61]: That data would be transmitted to the main control room in AKABEL through FOC with VSAT systems as backup

Commented [LMS62]: Please note that within the plant in AKABEL, LAN is being provided by TCC. LAN required at each gas manifold to be considered in EPIC scope

Commented [H63]: Noted

11.3. Voice and Data Communication

11.3.1. Voice Communication System

A voice-over-IP based telephone exchange system shall be provided in order to provide system wide telecommunication services between Central Control Room and all the stations.

11.3.2. Office LAN/ WAN

Structured cabling system (cabling and associated hardware) shall be provided in the manifold RTU's for data and voice communications. The Office LAN shall be completely separate from the Security LAN and process control LAN.

The telecommunication infrastructure shall be prepared to serve as future Office WAN. The dark fibers terminated in the relevant telecommunication boxes shall be secured together with the space for the future WAN equipment by others.

Commented [LMS64]: Please note that within the plant in AKABEL, LAN is being provided by TCC. LAN required at each gas manifold to be considered in EPIC scope

Commented [H65]: Noted

11.4. Security

11.4.1. Access Control System (To be discussed with Employer)

Electronic Access Control System is not envisaged for the manifolds, rather the access control will be effected manually using guard personnel.

Commented [LMS66]: Please note that within the plant in Akabel, Access Control System is being provided by TCC. Access control system in each manifold to be considered in EPIC scope

Commented [H67]: Noted

11.4.2. Intrusion Detection System

A perimeter control system shall be installed to detect any attempt to enter the manifold area without the required permissions.

The manifolds shall have at least two of the three different intrusion detection technologies, microwave, optical sensor cables and Infrared detection sensors installed.

The system shall be able to detect attempts to sabotage the system. In case of suspicious events an alarm shall be raised, and surveillance cameras shall be directed to the area in question. Optionally the station lighting in that area shall be switched on. Guards shall be provided with (prepared) action scenarios, escalation procedures and contact lists. The main central system shall be located in the security cabinet at AKABEL to be provided by others. Integration of the manifold's local system with the main AKABEL stations system shall be fully developed in detailed engineering.

Commented [LMS68]: Please note that this system is being provided by TCC in main plant in Akabel. System in each gas manifold to be considered in EPIC scope

Commented [H69]: Noted

11.4.3. CCTV System

CCTVs as video surveillance systems shall be installed at each manifold. The CCTV system design shall enable online and real time monitoring and recording of CCTV outputs both by the stations and from the security operation center at AKABEL. The main central system shall be located in the security cabinet at AKABEL to be provided by others. Integration of the station's local system with the main AKABEL stations system shall be fully developed in detailed engineering.

Commented [LMS70]: Please note that within the plant in Akabel, CCTV System is being provided by TCC. CCTV system in each manifold to be considered in EPIC scope.

Commented [H71]: Noted

11.4.4. Integrated Security System

All components of the CCTV and Intrusion Detection network are communicating to each other via dedicated Ethernet network. All Ethernet network components (e.g. Ethernet switches, media converters, routers and firewalls) delivered in the scope of the Integrated Security System (including the security subsystems CCTV and Intrusion Detection) shall be managed under the same NMS as the network equipment installed under the scope of the Data Transmission System.

The NMS provision shall be further discussed with employer and fully developed in detailed engineering.

The Integrated Security System in the manifolds will be tied into the main Integrated Security System in Akabel to be provided by OTHERS.

Commented [LMS72]: Please note that within the plant in Akabel, Security System is being provided by TCC. System in each manifold to be considered in EPIC scope.

Commented [H73]: Noted

12. Electrical Design Basis

12.1. Power Generation and Distribution System

Main power source for remote Manifold stations shall be via either of the following options or a rational mix:

- a. Through an underground high voltage transmission cable laid from AKAB Gas Processing Power Plant to the Platform Manifold Stations including AKAB, utilising step-up/step-down transformers & RMUs.
- b. Off grid solar/inverter power system designed to cater for the entire electrical, instrumentation, mechanical & process load of the facility and a standby generator which can either be methanol, gas, or diesel provided as emergency back-up with auto start/stop and manual override.

If option 2 is selected, the generator shall be able to startup and restore power within 15 – 20 sec of loss of power from the solar inverter system.

The Gas engine generator fuel can be either raw gas or Methanol pending final evaluation of the Power selection feasibility studies. If diesel generator is selected, 14-day autonomy diesel storage tanks shall be designed and installed in the remote Manifold stations for diesel supply.

Electrical load shall comprise mainly of lighting and small power (convenience) & welding receptacles, while Instrumentation & Telecom load shall comprise motor operated valves (if required), telecom racks and cabinets, electronic based monitoring, and control system equipment.

Mechanical/process load shall comprise mainly of sump/drain pumps if required, instrument air package and other equipment.

The power distribution system of the Project can be summarized as follows.

400VAC, 3ph, 50HZ for pumps, MOVs, and other package equipment

230V AC, single phase, 50HZ, for Lighting and Small Power (convenience & welding receptable) Loads.

Electrical Loads/equipment shall be supplied directly from either the 400/230VAC Main switchboard or dedicated distribution boards. Single line diagram detailing connected load shall be developed and issued.

The broad load will be approximately 18 – 22 kw per manifold, details provided separately in the Electrical discipline deliverables

Commented [AJ74]: Broad Loads shall be indicated. Approximate load shall be indicated.

Commented [H75]: See electrical load list

12.2. System Voltage and Frequency

The system voltage and frequency shall be 400/230 VAC and 50 Hz respectively.

12.3. Design Requirements

The main Electrical Equipment/Installation required for the Project includes

- Normal lighting and convenience outlets around the manifold platform area.
- Electrical pump for evacuation of drain pit at the stations
- Diesel transfer pump (if diesel genset is selected)
- Instrument air package
- Electrical cabling
- Facility Earthing and Lightning Protection System
- Tie-in to AKAB Complex's power plant & earthing system

12.4. Earthing and Lightning Protection System

Following the recommendations of IEC 61000-5-2 a single integrated earthing system shall be installed to reduce and control voltages to an acceptably low level for each Manifold station:

- Electrical Safety
- Lightning and Static electricity protection
- Intrinsic Safety

In addition to the earthing system, a lightning protection system shall be provided for protection of the entire facility from lightning discharge. Lightning air terminal shall be mounted on the roof of electrical room to provide lightning protection zone coverage for the entire facility.

12.5. Earthing of Plant Equipment and Structure

The metallic enclosures of electrical equipment shall be bonded to the plant earth grid. The metallic enclosures of non-electrical equipment (e.g. vessels) shall have a combined resistance to the general mass of earth not exceeding 4Ω , by means of one of the following:

- Bonded to the plant earth grid.
- Provided with their own duplicate earth electrodes.

Plant earth grid conductors shall have a cross-sectional area of 70mm^2 with reference to section (3.2.4) and (3.5) of DEP 33.64.10.33-Gen, for cross-sectional areas of branch conductors connecting equipment and structures to the Plant grid.

All earthing terminations shall be made with compression cable lugs. Interconnections shall be directly made using thermoweld or compression branch connectors. Thermoweld connection is preferred and shall be used wherever possible.

12.6. Area Classification

Area Classification will follow the provision of Shell DEP 80.00.10.13-Gen. which specifies requirements and gives recommendations for the determination of hazardous (classified) areas.

Commented [AJ76]: lightning protection system shall be included,

Commented [H77]: Noted/implemented

facilities handling flammable fluids and combustible dusts. The DEP is an amendment / supplement to IP 1.5, and comes with best practices for electrical equipment placement related to hazardous (classified) locations in Petroleum facilities.

Method will follow the process of analysing and classifying the environment where explosive gas atmosphere or combustible dusts may occur so as to facilitate the proper selection and installation of equipment to be used safely in that environment, considering the probability that explosive gas atmospheres can be present and the ignition characteristics of such gas.

Applicability

A facility that handles flammable and combustible fluids shall be classified in accordance with API 505 as amended and supplemented by Part III of the DEP or the installation of this project "ZONE" method shall be used for hazardous area classification.

12.7. Lighting

Normal and emergency lighting shall be provided for the stations via dedicated panels. Emergency lighting shall be same kind of lighting with the normal lighting but shall be integral with 3hrs battery autonomy. The installation shall generally follow IEC Standards and DEP specifications.

The lighting system shall be fed from a 400/230VAC photocell-controlled lighting distribution panel supplied from the main switchboard.

The illumination level shall be according to the requirements of referenced IEC Standards, and the Shell DEP Manual.

Lighting fixtures/luminaires shall be selected in consideration of suitability for hazardous area classification and ingress protection requirements.

Location and positioning of luminaires shall be done with the aim to adequately illuminate the entire platform area at BBOU, ELEG and the Pig Receiver area at AKAB respectively.

Lighting circuits shall be wired to separate junction boxes.

12.8. Lighting Poles

The lighting poles shall be fabricated from hot dip galvanized steel plate with octagonal cross section. The nominal height above ground level to the centre line of the fitting shall be 8 meters in the plant area and 3.5 meters in the valve platform area.

The flood lighting poles shall be of the collapsible type of Abacus design or equivalent.

The essential structure and dimensions of the poles shall be suitable to accept one mounted fixture. Due regard shall be given to corrosion, rust proofing and final paint finish for the stated climatic conditions.

An earth terminal stud shall be provided on the outside base of the pole for lugging and termination of ground cable to plant earthing grid. The lighting pole shall be equipped with a mounting plate for installation of external lighting junction box. Reference shall be made to Shell standard drawing S.69.001 and S.69.003.

12.9. Cables and Cabling System

All cables shall be in accordance with the DEP and the Project Specifications. All cables shall be at least flame retardant in accordance with the requirements of IEC 60332- part 3. Cables for safety related systems such as fire & gas detectors, emergency/fire push-buttons, CO2 release, etc shall be both flame retardant and fire resistant in accordance with IEC 60331.

Cables shall be armoured and mechanically protected against mechanical shocks.

Electrical cables for power and control shall be designed as indicated in DEPs.

Cables shall be either of multi-strand, round copper conductors, XLPE/SWA/PVC or single core stranded copper conductor, XLPE/AWA/PVC type.

12.10. Cable Routing

Cable runs shall as much as possible, be designed considering aesthetics and cost effectiveness. The cables on the platforms shall run on cable tray as per Shell DEPs.

Within the battery limit at AKABEL, all low voltage power cables shall be direct buried at a depth of 600 mm (min) in cable trenches, except at road crossings where cables shall be installed through a PVC pipe to avoid road damage in case of cable repair and replacement work.

Aboveground earthing cables shall be PVC sheathed, coloured yellow/green. The PVC sheath is a protection against electrolytic corrosion. Generally, underground earthing cables should be of the same type. However, where the requirements of this DEP with respect to earth resistance values (5.5.1) or EMC (5.5.6) cannot be met, underground earthing cables may be bare.

12.11. Evacuation Pump

The power to the evacuation pump to be installed at the stations shall be fed from the respective LV Switchboard including AKAB station whose power supply is from the nearby power plant in the methanol processing plant.

12.12. Accepted Electrical Equipment Vendors

Shell's material approval for all electrical vendors shall be followed during the procurement and shipping of electrical installation items to the site. Reference shall be made to the document – "SPDC List of Accepted Electrical Equipment Vendors."

12.13. Testing and Commissioning

Suitably qualified personnel shall carry out equipment testing and commissioning in line with DEP 63.10.08.11-Gen (Field commissioning and maintenance of electrical installations and

equipment). All installation and testing equipment to be employed for the installation activities shall have valid certifications. All test results shall be recorded on standard test sheets and shall be approved by the Client.

13. Civil / Foundations Basis Of Design

13.1. Civil Scope of Work

The Civil/Structural Design scope for this project includes the following:

- Structural/Foundation design for station platforms at BBOU, ELEM and AKAB
- Structural/Foundation design for Monorail at BBOU, ELEM and AKAB station
- Structural/Foundation design for Pipe Supports at BBOU, ELEM and AKAB
- Structural/Foundation design for Valve Supports AKAB
- Drain Pit design at BBOU, ELEM and AKAB
- Pavement Design at AKAB

13.2. Geotechnical Data

The Geotechnical Data required for all Foundation designs on this project shall be based on the Site-Specific Geotechnical Reports for the manifold stations at Boubouwei, Elepa and the plant at AKABEL. The reports shall supply recommendation for foundation types based on type of structures to be borne by the foundations

The Geotechnical Data required for all Foundation designs on this project shall be based on the Site-Specific Geotechnical Reports for the manifold stations and the plant at AKABEL. The reports shall supply recommendation for foundation types and other required information necessary for the Project.

13.2.1. Soil Type

Sub-Soil Investigations along the pipeline right of way at the depth explored indicate that the soil is mostly underlain by dark grey soft peaty/silty clay with low bearing capacity.

13.3. Environmental Data

See section 3.0

13.3.1. Entomology and Vermin

As prevalent in the Niger Delta, with thick forest and swamp habitat.

13.3.2. Fungi and Growths

As prevalent in the Niger Delta, with thick forest and swamp habitat.

13.3.3. Lightning

Severe & frequent in the rainy season (6 months in a year).

Commented [AJ78]: EPIC to provide load intensity of structures. The load intensity depends on the load per column supporting platform for which, number of column proposed for platform to be shared with Brone.

Commented [H79]: This document is BoD and as such this detail will be provided in the requisite discipline deliverables

Commented [AJ80]: Geotechnical Report has been reviewed by Epic and Epic have shared their comments.

Commented [H81]: EAL comments have been hinged on paucity of geotechnical input data from geotechnical report

Commented [AJ82]: Geotechnical Report has been reviewed by Epic and Epic have shared their comments.

Commented [H83]: Same as above

13.3.4. Site Altitude

The Site Altitude to be referenced to the Mean Sea Level (MSL).

13.4. Software

The following computer software shall be used:

- Staad Pro for Analysis of Onshore Structures
- Microsoft Excel and Microsoft word for Preparation of Civil and Structural Design Calculation reports.
- AutoCAD for Drawings

13.5. Units of Measurement and Unit Weights

See section 1.6

13.6. Material Unit Weight

Table 11.0: Material Unit Weight

S/N	Material	Description	Unit Weights	Unit
1.	Concrete	Reinforced and Plain Concrete	25	KN/m ³
2.	Earthwork	Soil/Sandy laterite	18-21	KN/m ³
3.	Steel	Steel (General)	78.50	KN/m ³
		Grating	0.44	KN/m ²

13.7. Material Quality

Table 11.1: Material Quality

S/N	Material	Description	Strength	Unit
1.	Concrete	Reinforced	25/30	N/mm ²
2.	Steel	Rebar	410	N/mm ²

Commented [AJ84]: Grade of Concrete is mentioned as M 25 in 13.7 whereas in 13.8.9 it is mentioned as M30

Commented [H85]: Range of M25/30 for respective foundations

13.8. Civil Structural Design Philosophy

13.8.1. Datum

All levels to be used in design shall be given relative to established site datum as indicated on the

plot plan drawings.

13.8.2. Design Life

Major elements of the civil structures shall be designed for a minimum effective life of 30 years.

13.8.3. Maintenance Philosophy

Structures shall be designed to minimize maintenance requirements wherever practicable. Single robust structural steel elements are preferred in lieu of multi-member elements. All equipment and fittings shall be mounted, and access provided to facilitate in-service inspection and maintenance.

13.8.4. Durability of Materials

All materials shall be of approved types conforming to the relevant international and national standards. All materials shall be accompanied by the appropriate testing, calibration, and certification documents.

13.8.5. Structural Steel Work

All structural steel shall be grade S275 or S355 to EN10025 (Hot Rolled Products of Non-Alloy Structural Steels) and designed to meet durability requirements of relevant codes and standards, refer to the project document - Structural Steel Specification.

13.8.6. Reinforced Concrete

Reinforced concrete construction shall be designed to achieve high-grade strength, workability, and durability appropriate for the environmental conditions of exposure on site during its service life.

All reinforcing steel and concrete materials to be used shall conform to requirements of relevant Codes and Standards.

Refer to the project document -Specification for Concrete Works.

The minimum concrete cover to the outer reinforcement shall be as recommended in EN1992 Design of Reinforced Concrete Structures and a minimum blinding layer of 50mm thickness shall be placed under all shallow foundation.

13.8.7. Foundation Type

Considering the low bearing capacity of the soil, the foundation type to be adopted will be pile foundation and where soil capacity is improved to bear incoming load, Isolated footing may be considered.

Pile factor of safety = 3.0

Piles may be either concrete or steel or both, with their selections based on the expected loads, location, availability, cost and general site conditions.

Commented [AJ86]: EPIC to provide load intensity of structures. The load intensity depends on the load per column supporting platform for which, number of column proposed for platform to be shared with Brone.

Commented [H87]: See response to comments in Sec 13.2

13.8.8. Concrete Cover

Based on the geotechnical investigation report on the chemical nature of the soil and the water level in the site, the reinforcement bars shall be placed with a minimum cover of 75 mm in ground beams, and any other concrete member in direct contact with the ground. Everywhere else a minimum of 40 mm cover will be used.

Based on the geotechnical investigation report on the chemical nature of the soil and the water level in the site, the reinforcement bars shall be placed with a minimum cover of 50 mm in ground beams, and any other concrete member in direct contact with the ground. Everywhere else a minimum of 30 mm cover will be used.

Commented [AJ88]: Concrete cover shall be 75 mm for structure coming in contact with soil and others shall be 40 mm

Commented [H89]: Noted

13.8.9. Concrete Grade

Reinforced concrete shall be grade 30 for heavy load bearing foundations the strength shall be verified by cube tests carried out regularly on samples obtained from different concrete batches and the results shall be made available during construction phase.

The minimum grade of reinforced cement concrete used shall be M30 for all concrete structures including foundations.

Based on recommendations from the soil investigation report, the use of higher cement content and/or specific type of cement shall have precedence where necessary.

Reinforced concrete shall be grade 30, the strength shall be verified by cube tests carried out regularly on samples obtained from different concrete batches and the results shall be made available during construction phase.

The minimum grade of reinforced cement concrete used shall be M30 for all concrete structures including foundations.

Based on recommendations from the soil investigation report, the use of higher cement content and/or specific type of cement shall have precedence where necessary.

Commented [AJ90]: Grade of Concrete is mentioned as M 25 in 13.7 whereas in 13.8.9 it is mentioned as M30

Commented [H91]: Grade of concrete is a range of M25 – M30 for respective structures

13.8.10. Reinforcement Bar

A single steel grade will be used throughout which will have a minimum yield strength of 460 N/mm². Bending shapes and standard bar diameters will be according to BS 8666:2005.

13.9. Structural Analysis and Design

13.9.1. General

The design considers not only the properties of the materials and the calculated stresses for the members, but also the prevailing conditions of the local environment and the requirements at the site, and their effect on cost to achieve a safe and economic design.

Structural steelwork shall be designed in accordance with the requirements of EN 1993 while

reinforced concrete for the works is designed in accordance with the recommendations of EN 1992, EN 206 and ENV 13670.

13.9.2. Analysis and Design

The analysis of both reinforced concrete and steelwork shall be elastic analysis to determine the distribution of forces within the structure and where appropriate computer-based software shall be used for rigorous analysis.

The design of both reinforced concrete and steelwork shall be to Limit State method of design (Ultimate Limit State and Serviceability Limit State).

13.9.3. Load Combinations and Load Factors

Load Combinations and Load Factors for Ultimate Limit State design are in accordance with the following:

- EN 1990 – Basis of Structural Design
- EN 1991 – Action on Structures
- EN 1992 – Design of Concrete Structures
- EN 1993 – Design of Steel Structures respectively

And consider the design situations, all critical and adverse loading conditions that the structure may be subjected to during installation and in service.

13.10. Super Structural and Sub structural Design

13.10.1. Super Structural Design

Wind pressures on super structures shall be determined based on the basic wind speed over the site and applied for the determination of wind loads on the main vertical steel structural members.

Axial loads on Superstructures shall consist mainly of equipment variable, permanent and super structural self-weights. Members shall be checked for their adequacy against buckling, compression, and shear failures.

The normal condition situation shall be adopted as the design condition on the request of the customer.

13.10.2. Supports

Pipes must be adequately supported both vertically and horizontally. Support must be designed such that free movement due to temperature fluctuations is allowed.

Support shall be provided near fittings and joints. Additional support shall be provided at branches, at valves, at heavy fittings and elsewhere on site where conditions dictate. Support must consist of hangers, pipe clamps, rollers, guide sleeves etc, depending on the condition on site

13.10.3. Foundation Sub-Structural Design

Foundation designs shall be carried out to EN1997-Geotechnical Design, using engineering properties, results and parameters obtained from the Geotechnical & Site Investigation Report.

The method(s) to be used for demonstrating that limit states will not be exceeded shall be identified by the design Engineer and in compliance with EN 1997 & EN1992.

All designs shall be made with due allowance for safety, constructability, costs, programme, reliability, risks, performance and practicality of the structures in mind.

Foundation design on this project shall be in accordance with the project specification for foundation works. Shallow foundations such as pad footings may be used to support some of the lightly loaded structures and equipment. Tops of foundations shall be a minimum of 300 mm above grade and shall be higher if required by flood conditions. Footings shall not be located in an area shown by the subsurface investigation to be traversed by underground water or seepage unless these conditions are remedied. Minimum cover to reinforcement shall be 50 mm for concrete above grade. Minimum cover to reinforcement shall be 50 mm for concrete below grade. Minimum foundation blinding thickness shall be 75 mm.

Foundations shall be designed to have a minimum safety factor of 1.5 against overturning and 1.5 against sliding. Apart from a detailed check against excessive settlement all foundations shall be checked for adequacy against bearing capacity, bending, and punching and shear failures.

The loading considered in the foundation design shall include.

The maximum reactions obtained from Structural Analysis of the Super Structural Support frame work.

The maximum shear from primarily wind loads and tensile loads in the super structural Steel Frame Work at the level of Concrete Foundation.

13.11. Foundation Design

Foundation design shall be carried out using engineering properties, results and parameters obtained from the site investigation and geotechnical investigation reports. The foundation shall be designed in accordance with all relevant BS EN codes especially BS EN1997 to withstand loads or stresses and to meet other serviceability requirements during handling, installation and service.

Since part of the pipeline right of way will be excavated and sand filled, the bearing capacity for shallow foundation shall be designed for a bearing capacity based on the geotechnical report

13.11.1. Pit Design

In general, Pit bases shall be designed at minimum of 1.0m depth from grade level.

Commented [AJ92]: Minimum foundation blinding thickness shall be 75 mm.

Commented [H93]: Noted

13.11.2.Design Considerations

The recommendations of EN 1992 shall be adhered to unless otherwise specified.

13.11.3.Loading Assumptions

Design of concrete pit shall include consideration of the following loads:

- Permanent weight of pit
- Surcharge and external Soil Bulk Pressure on Wall should be checked without water pressure acting on the inside of the walls.

13.11.4.Design Criteria

Pit Wall Design shall be in line with American standard codal requirement and shall consider the following:

- When the length of the wall is more than three times its height, the wall shall be designed as a vertical cantilever.
- When the length of the wall is equal to or less than three times its height, the wall shall be designed for two-way bending.
- When the wall is relatively rigid or has a top slab, the wall shall be designed using earth pressure at rest coefficients. Backfill around walls shall normally be selected granular material. Other backfills allowed for in the Detail design may be used. The type of backfill shall be noted on the foundation drawing. Reinforced concrete walls should be a minimum of 250mm thick reinforced on each face and designed as water retaining. 200mm may be used for small/shallow pits. Minimum wall reinforcement, vertical and horizontal, shall be in accordance with the requirements of the specified codes.

Bottom Slab Design shall be designed as either a one or two way spanning slab. Moments transferred from wall into slab should be included. Minimum slab reinforcing shall be that required by the relevant code of practice.

The minimum slab thickness should normally be 250mm with two layers of reinforcing placed in the slab. The area of top steel shall be equal to the area of bottom steel. 200mm may be used for small/shallow pits. A 50mm thick blinding layer shall be provided beneath the slab.

13.11.5.Upthrust from Hydrostatic Pressure

The pit shall be designed for a minimum safety factor against upthrust of 1.1 for empty conditions.

13.11.6.Fencing

The total plot area shall be fenced including the areas designated for future expansion. The fencing requirements and specification (including gates) should be in line with SHELL's security guidelines. The security fence shall be high security steel mesh of 4.5 m height all around the perimeter of the plot plan. A Surveillance walkway shall be provided around the fence.

Commented [AJ94]: Pit Wall Design shall be in line with American standard codal requirement

Commented [H95]: Noted

13.11.7. Pavement Design

Pavement Design shall be done in line with codal requirement.

All roads shall be heavy duty, and will be designed based on recommendations from the American Association of state High Way Transportation Officials (AASHTO). The minimum road width shall be 4m. All paving is to be sloped in compliance with the requirements for drainage. Culverts shall be considered part of the road system.

The traffic and loading information required for the design of road pavement includes:

The equivalent single axle load of 150kN and 50kN on heavy and light duty roads.

Traffic loading shall be specified based on the recommended number of axle movements as in Table below

Table 11.2: Traffic Loading

S/N	Roads	Loading
1	Heavy- duty roads	1.5 millions
2	Light- duty roads	1.0 millions

California Bearing Ratio (CBR) shall be taken in line with soil investigation report for the design.

The CBR of sub-grade materials shall be at least 10.

Concrete for paving construction shall be in accordance with required subgrade preparation and allowable soil bearing recommendations per geotechnical investigation and report. For light duty paving, the minimum concrete slab thickness shall be 100mm, and the minimum reinforcement of one layer of WWF, material ASTM A185 in accordance with ACI 318. For heavy duty paving, the minimum concrete slab thickness shall be 200mm. Concrete for paving shall not be less than Grade C30.

The minimum requirements for paving type are as follows:

Table 11.3: Paving Type

Paving Type	Thickness	Reinforcement	Maximum Expansion Joint Spacing	Maximum Contraction Joint Spacing
Light Duty	100	1 Layer A193	20mm	20mm
Heavy Duty	200	2 Layer A193	20mm	20mm

Commented [AJ96]: Pavement Design shall be done in line with codal requirement before arriving thickness of 100mm and 200mm CBR shall be taken in line with soil investigation report

Commented [H97]: Noted

14. Operations and Maintenance

14.1. General

Operations activities shall be categorized into core and non-core activities which are necessary for efficient management of the pipelines and the manifolds.

The core activities include:

- Operations management
- Production operation including frontline maintenance
- Production measurement, allocation & accounting
- Asset management
- Production planning
- Operations input to projects

Non-core activities (which are support activities that are either temporary or specialized or that require minimal skills) include:

- Pipeline pigging
- Laboratory services
- Logistics operations
- Heavy lifting using cranes
- Supply of chemicals and other consumables
- Specialized major maintenance
- etc.

14.2. Process Control and Safeguarding System

Process control and safeguarding system shall be provided for the pipelines for the purpose of monitoring and protecting the pipelines from adverse operating conditions that could result in harm to people and/or damage to the asset or environment.

Pipelines control and monitoring systems shall be designed to be fail safe. Control and monitoring shall be done from the Gas Plant Control Room.

Field instruments should be capable of handling digitally transmitted calibration and maintenance information (remotely calibrated and tested). The main control system should provide a means to monitor, diagnose problems, and perform surveillance remotely.

- Flow, Temperature and Pressure control system shall be installed on the pipelines for monitoring and control purposes.
- Transmitters shall be installed to transmit information on the parameters to the Control room.
- The field elements and the transmitters shall provide information to the Operators in the

Control room for quick response in order to ensure the pipeline is operating within the design envelope.

- The Process safeguarding system shall be independent of the process control system.
- The process safeguarding system shall ensure that suitable protection is provided against the maximum pressure (and other operating parameters) that can be generated by the worst credible malfunction.
- Control valves forming part of the process safeguarding system, should not have a bypass, unless the bypass is locked closed.

14.3. Maintenance Support Function

The maintenance function will be organized as part of the Operations function to perform regular predictive, preventive, and corrective maintenance activities.

Maintenance support for Field Operation shall include:

- Workshop provided by BFPCL
- Valves repairs, calibration and servicing
- Inspection
- Provision of maintenance manpower supply
- The details and types of maintenance support services should be reviewed and agreed between the service providers and service users, mainly the field Operations teams. They should be equipped and manned to handle all repair works that cannot be done on site. The scope would also include minor modifications and fabrication works.
- The maintenance support function will hold manpower call off contract(s) for skilled and semi-skilled personnel for provision of the above Maintenance Support services

14.4. Field Logistics and Transportation

The gas well heads, satellite manifolds and evacuation manifold with operational valves are located in swampy terrain. Regular visits to these manifolds will be required for operation and maintenance activities. Therefore,

- Operators shall carry out its field operations in a manner that will minimize or eliminate risk to personnel, environment, and assets. Safe and efficient means of transportation shall be provided at all times, in all situations, and under every condition.
- The field logistics philosophy shall cover transportation of personnel and materials into, within, and out of the fields.
- All logistics supports will be provided by BFPCL or through one of its qualified, competent, and experienced service providers.
- All routine material logistics support will preferentially be by marine transportation.
- Movement into, within, and out of fields shall be by marine transport with company provided

boats (owned or leased) under armed security escort.

14.5. Health, Safety and Environment (HSE)

- a. Health, Safety and Environment (HSE) shall be an integral part of BFPCL Gas Gathering Pipelines Operations.
- b. Operations HSE Cases shall be prepared by the Project team and handed over to Operations as part of Project Hand Over and implemented/owned by Operations.
- c. All HSE critical roles in Production Operations shall be performed by certified competent staff. A competency assurance process will be developed for implementation to ensure that this is met.
- d. Contractors providing services for the facility shall develop HSE plans, setting out performance targets and methods to be adopted to achieve set targets.
- e. Operations activities shall be carried out in compliance with BFPCL HSE policy. The objective will be to assure that the health of employees, contractor personnel and others working in the facility will not be adversely affected by the work.
- f. A PTW (Permit to Work) system shall be implemented for all non-routine work which includes maintenance and repair, construction and modification, hot work, entry into confined spaces, work in exposed location, pressure testing, heavy lifts, etc.
- g. The above listing on PTW shall include activities where and when safeguards may be temporarily bypassed, or when new hazards may potentially or inadvertently be introduced, e.g., hot work in a producing facility, or in an area where there may be the presence of hydrocarbons, Pipe breaking, etc.
- h. Selection, approval, and procurement of chemicals will be in accordance with BFPCL approved procedure.
- i. Chemicals must be acquired, transported, used and disposed of in compliance with NUPRC, NMDPRA and FME regulations.
- j. No new Chemicals will be accepted from a supplier or vendor without the appropriate Material Safety Data Sheet (MSDS).
- k. All chemicals will be covered in the Safe Handling of Chemicals (SHOC) cards and enforced through the installation and operation of the facility.
- l. All personnel will be prevented from exposure to noise in amounts, which may cause noise induced hearing loss, or may affect employee wellbeing.
- m. Noise contours and maps shall be developed at the Operate phase of facilities.
- n. In areas with noise levels above 85-dB (A), the use of hearing protection shall be mandatory.
- o. Environmental Impact Assessments (EIA) shall be conducted pre and post projects in order to identify baselines and evaluate potential impacts of any project and to minimize negative impact on the existing environmental conditions.

Commented [AJ98]: Please refer to the comments furnished on Operation & Maintenance Philosophy document and update it accordingly.

- p. The facilities shall be designed, operated, and maintained to meet stringent environmental protection objectives within the boundaries of economic, technical, and social constraints.
- q. The emission and discharge levels from BFPCL operations (such as emission/waste lube oil from generators, if any) shall be routinely monitored against regulatory limits.
- r. All fluids disposed into the environment shall be treated to the prescribed quality prior to disposal.
- s. Flaring and venting of gas (e.g. from pigging, PSVs, flarestacks) shall be eliminated under routine operations or as a minimum kept to an absolute minimum, within applicable guidelines and regulatory requirements.
- t. A Waste Management System will be put in place prior to commissioning and will document all waste streams, sources, segregation approach, and disposal methods.
- u. All domestic and industrial wastes (especially fluid waste from pigging) generated shall be collected, segregated, and disposed of onshore.
- v. The Operations Manual for all producing facilities shall contain Emergency Response Procedures covering foreseeable emergency situations (e.g., fire, spill, loss of containment, medical evacuation, security incidents, etc.)
- w. The Emergency Response Procedure shall be duly signed, regularly reviewed and updated, communicated effectively throughout the organization, including pasting in appropriate places.
- x. There shall be regular periodic drills involving all cadres of the organization.
- y. To improve the emergency response capability in the field, formal emergency training and drills are to be periodically carried out. Recommendations and lessons learned shall be built in for improvement.

14.6. Field Security

- Security considerations at all times shall cover personnel, asset, and property; in this order of priority.
- Security measures must be pre-emptive/proactive and not reactive. Therefore, robust and fit-for-purpose security assessment and plan shall be developed prior to commencement of any field operation and activity and must be implemented accordingly.
- Contractors shall be responsible for arranging and managing their own security in accordance with the rules of engagement as issued by the BFPCL Security organization.
- Evacuation of personnel from a facility or Field Logistic Base (FLB) under security threat must be adequately planned and provided for as part of normal operation. Evacuation plan will be prepared and drills carried out to ensure familiarity with the procedure.
- Armed escort will be provided to accompany and work with teams if they have to travel and work in high security risk locations.
- Field security plan shall be built around local intelligence gathering and use of Government

Security Forces (GSF) for enforcement.

- Management shall be responsible for the provision of resources to implement the security plan. Field Management shall be responsible for resource definition and utilization to implement field security.
- Contractors / vendors engaged at field shall submit an acceptable security plan which shall seamlessly tie into the field security and the State security plans.
- In the event of community disturbance and/or illegal tampering, the facilities shall be made safe through automated failsafe shutdown or remote shutdown initiation from the control room by Production Operations personnel.

14.7. Field Communication

Reliable communication network is critical to ensure a safe, reliable and efficient operations and during an emergency situation. An integrated telecommunication network shall be provided for the support of integrated nodal operations and establish data linkages with other associated stakeholders. The following strategies will be adopted:

- Field communications will be based on maximum use of reliable commercial communications infrastructure, where available.
- Telecommunication system shall cover all the necessary facilities such as telephones, mobile radio, CCTV, and data highways (ICSS for control and monitoring purposes).
- Access control systems (ACS) and Pipeline intrusion detection system (PIDS).
- A public address system (PAGA) within the local control rooms shall be provided to enhance site communication and emergency response preparedness.
- Email/Internet/: Data, IP Cameras/Video Conferencing should be provided. Scope of coverage will include all facilities and locations that have a direct impact on operations objectives.
- Field locations will be linked to the main Communications hub by Fibre Optic Cable (FOC), microwave radios mounted on masts/towers will be used as secondary communications link.
- Fibre optic cables shall be laid together within new pipelines right of way for primary communication link.
- Access to LAN (Local Area network) and WAN (Wide Area Network) shall be available at all field locations via the FOC from the base communication system.
- Approved users will be able to use laptop computers to access emails, web-sites and other applicable systems as deployed.

14.8. Operating Strategy for Field Manifolds

The two gas gathering manifolds for phase 1 of this project are the BBOU and ELEP manifolds. The BBOU manifold and ELEP manifold will not be manned but will be attended for operations and maintenance purposes.

Access shall be provided to the manifolds through the water ways from the gas plant to allow personnel travel from the plant location to the manifolds for operational reasons by boat.

The manifold shall have elevated platforms with perimeter fencing to prevent unauthorized entry to the platforms.

To ensure the security of the valves and equipment on the platforms, a surveillance team shall be engaged from the community, to be supported by the security agents. Appropriate monitoring technology shall be considered to provide advance warning of intruders.

Communication system shall be provided between the platforms and the plant control room to facilitate passage of information to/from the locations.

Motor operated valves shall be designed / installed on the manifolds for easy operation. Pig launcher and receiver shall be installed on the manifolds as part of pipeline design for regular pipeline cleaning.

Primary Power supply for the manifolds shall be through power cable from Akabel. Consideration will be given to other back up power system e.g, solar or diesel generator, to ensure higher reliability.

Shelter shall be provided on the platform to protect personnel from weather elements.

14.9. Hydrocarbon Measurement & Accounting

All incoming and outgoing hydrocarbon streams to and from field facilities shall be measured, using appropriate technology and practice to meet contractual, production, fiscal, custody transfer, royalty, and regulatory requirements as per the measurement design philosophy.

Appropriate hydrocarbon accounting procedures shall be established to cater for present and all future evacuation processes.

Specific requirements and details will be advised after the final conclusion of the GSPA.

Commented [AJ99]: Narrative is generic in nature. Epic should furnish the Project Specific Requirements & Details.

14.10. Metering System

The overall production measurement philosophy shall be developed and agreed as part of the Gas Sales Agreement.

Gas measurements for fiscal / custody transfer, allocation, and production measurements shall be in accordance with the respective NUPRC/NMDPRA guidelines for gas and liquid measurements.

The NUPRC/NMDPRA preferred / approved meter type for fiscal / custody transfer measurements for liquid are Coriolis meters and others while for gas is Daniel Orifice.

Other meter types (V-cone, Ultrasonic, etc.) may also be considered, however, an approved waiver shall be sought and obtained from NUPRC/NMDPRA.

All fiscal and allocation metering stations shall be designed as a dual stream facility to ensure maximum availability. Thus, one metering stream is always available to enable continued gas export metering during routine maintenance, inspection and validation of any of the metering facilities.

14.11. Incoming Gas Stream

Incoming raw gas stream from each SPDC Manifold will be metered at the Custody Transfer Point within the Manifold for billing, while a second meter shall be provided at the Gas Plant outlets for lean gas and condensate metering for pricing.

14.12. Condensate Export

Custody transfer metering shall be provided at delivery points of condensate export pipelines with a meter at injection points.

15. BOD Matrix

Situations & Systems			Design Basis	Remark / Reference
1	Location			
1.1	Gas Source		OML 33 / 35 - with locations widely spread & on both sides of Nun River	
	1.1.1	Target Fields	BBOU, ELEP, NUNR, SEIB, OPUG	
			BBOU & ELEP in OML 33 and NUNR in OML 35, - East of Nun River	
			SEIB in OML 35, OPUG in OML 35 and , - West of Nun River	
1.2	Gas Gathering scheme		Nodal Raw Gas Gathering thru Pipeline Network to ELEP, and from there to Akabel	
	1.2.1	Receiving Nodes	Four Nodes - ELEP, BBOU, NUNR & SEIB Nodes - geographically convenient	
			SEIB Production to SEIB Node S-M/F – geographical convenience.	
			SEIB Node inventory and NUNR Production to NUNR Node S-M/F - geographical convenience	
			BBOU & OPUG Wells Production to BBOU Node S-M/F - geographical convenience	
			ELEP Wells Production, BBOU Node Inventory, & NUNR Node Inventory to ELEP Node E-M/F - geographically convenience	
	1.2.2	Evacuation Node	ELEP Manifold	
			Inventory from ELEP Node E-M/F to AKAB PLEM, and from there to the	

			Gas Plant	
1.3	Gas Gathering System	A mix of Wellhead Flowlines, Field Manifolds, and Bulk Transport Lines, - optimally configured in terms of cost, layout, and choice of evacuation manifold (Elepa)		
		Wellhead Flowlines Gather Each Field Production into the Central Field Manifold (F-MF) within the Field, followed by:		
		Transportation of the F-MF inventory through Bulkline to Satellite Manifold (S-MF) / Evacuation Manifold (E-MF).		
		Transportation of the S-MF inventory through Bulklines to the E-MF		
		Transportation of the E-MF inventory through a Trunkline to the Gas Processing Plant.		
1.4	The Gas Processing Plant	The Gas Plant will be located at Akabel, Brass Island, Bayelsa State of Nigeria, near the shores of Atlantic Ocean.		The Gas Processing Plant is outside the scope of this FEED which is limited to the Gas Gathering Pipelines
		The Plant Site is in the same complex with the Methanol Plant and the Power Plant that it will be supplying the processed gas		
2	Business Case			
2.1	Justification	To supply processed gas needed as feedstock to Methanol Plant and a Power Plant, - in the value chain for methanol production by the methanol plant, and power generation by the power plant		
		GSPA with SPDC in place for the Supply of the Natural Gas to be Processed		

2.2	GSPA (Gas Sales and Purchase Agreement)		Wellhead (Raw) Gas at Each of the Fields' Manifolds for offtake by BFPC through own Bulklines	
	2.2.1	Gas	340 MMscfd net	
	2.2.2	Supply Pressure	80 - 100 barg	
	2.2.3	Condensate	11 – 34.4 Mpcd	
2.3	On-Stream Sequence		Phased Sequential (Production) Injection: BB & ELEP + NUNR + SEIB + OPUG in that order, - with stop points driven / moderated by the GSPA	

Commented [AJ100]: 340MMscfd would be Treated Gas at the outlet of GPP. Feed gas quantity should be commensurate to 340MMscfd at GPP outlet.

Commented [H101]: Noted

3 Gas Offtake / Flow Conditions

3.1	Flow Rates / Fluid Characteristics			NB: SPDC yet to issue Forecast; figures will be updated thereafter. 350 bpd is correct
	3.1.1	Gas Production	386 MMSCFD from year 1 to year 18.	
		Condensate Production	Up to 35,000 bpd	
		Water Production	Up to 350 bpd from the water content of the Natural Gas	5.35 Mbopd of Condensate
			Up to 16.5 Mbwpd reservoir production	Also water production will occur, starting at 1.7 Mbpd in year 1 and increasing to about 20 Mbpd in year 18
		CGR	7 – 161.7	
	3.1.2	Gas SG	0.6 – 0.74 (Air = 1)	
		Condensate SG	0.74	
		Water SG	1.00 – 1.02	
	3.1.3	Corrosivity Potential	Envisaged, on account of CO ₂ in the gas stream	Fluid therefore to be considered two phase since no treatment will be done before injection into the Line, - prompting design of the Pipeline for two-phase Handling
	3.1.4	Souring Potential	Not Envisaged, - since no H ₂ S in the gas stream	
	3.1.5	Foaming Potential	Not Envisaged	
	3.1.6	Scaling Tendencies	Not Envisaged	

Commented [AJ102]: This figure is taken from trewan Report, EPIC to clarify figure from Latest forecast to follow or Trewan Report to follow. Please check.

Commented [AJ105]: Check whether this figure is Correct.

Commented [AJ106]: 1)CGR ratio is matching with Trewan Report, But as per latest Forecast the figure is 4.35 to 67.57. EPIC to check it.
2) EPIC to check that to have uniform data throughout the document for Water and condensate flowrate.

Commented [AJ103]: This figure is taken from trewan Report, EPIC to clarify figure from Latest forecast to follow or Trewan Report to follow. Please check.

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				Two Phase Gas Transportation all the way to the Gas Processing Plant at Akabel Consequence is that Frequent Pigging will be needed to keep the Lines operating
3.2	Pressure Regime			Provision to accommodate higher pressure (up to 110 barg) at the distant field manifolds of Seibou, Opugbene
	3.2.1	Feed Stream Inlet Pressure	80 - 100 barg	
	3.2.2	Feed Stream Outlet Pressure	60 - 65 barg	
	3.2.3	Gas Plant Processed Gas Outlet Pressure	41 - 49 barg	
3.3	Product Metering		The NAG facility shall be provided with metering devices for gas and condensate flows. These meters shall be custody transfer type meters. All the instruments provided shall meet the Hazardous area requirements for field instruments	
	3.3.1	Raw Gas Metering	Venturi Meter at the Custody Transfer Points at the Field Manifolds	
	3.3.2	Feed Stream Outlet Pressure	60 - 65 barg	
4	Design Philosophy			
	4.1	Fitness for Purpose & Functionality	The project shall be designed to meet the operating capacity objective of 340 MMscfd at the Gas Processing Plant outlet and associated NGL and	Capability of transporting and processing the required volumes

Commented [AJ107]: Check with I&C.

			water production, allowing for planned maintenance & expected system reliability	of products
			Materials & Equipment will be selected based on Pressure, Temperature and Fluid Characteristics	
	4.1.1	Project Quality Delivery	Driven by an understanding of what constitutes a good (an excellent) design that will deliver technical & operational excellence; - requiring a good knowledge of how humans interact within the work system	
	4.1.1	Human Factors Engineering (Ergonomics)	Inclusion of User Requirement through: - Consideration to human capabilities, limitations and needs in the interaction between humans, technology and the working environment. - Equipment, Valves, Controls & Accessories Layout Design, degree of automation, and extent of protective systems.	Operability and Maintainability of the System to be given priority, considering the peculiarities of geographical locations of the Facilities. Balance will be met between automated and manual operation, and between ease & cost of maintenance
4.2	Mode		Not normally manned facility, with Automatic Shutdown Capability	Automatic Shutdown to be Fail Safe
	4.2.1	Automation	Remote Shutdown Capability from a PLC based Control Room.	Consideration to the use of Self Contained Fail Safe Shutdown System that Requires Human Intervention for

				Reopening
			Localised startup capability from the Field M/F & PLEM, Either as a provision, or intended startup process for the P/L	This will require the Installation of Patch Panels & Fieldbuses. For such a small installation, it will be more economical to keep automation as simple as possible
			Remote Terminal Unit with SCADA capability	
4.3	Governing Regulations		Oil Pipelines Act, Chapter 338, Laws of the Federation of Nigeria 1990 DPR (NUPRC) Guidelines for Installation of Gas Plant	
	4.3.1	Regulatory Approvals	PTS (Permit To Survey) OPL (Oil Pipeline Licence) EIA (Environmental Impact Assessment Study) Staged Approvals contained in DPR (NUPRC) Guidelines for Gas Plant Facility and Oil & Gas Pipeline Systems	
	4.3.2	Reference DPR (NUPRC) Guidelines	Guidelines for the Establishment of Natural Gas Plant Facility in Nigeria DRP P-1P: Guidelines & Procedures for the Design, Construction, Operation and Maintenance of Oil & Gas Pipeline Systems in Nigeria - 2007	
4.4	Codes & Standards			ISO 13623:2009 § BS EN 14161:200
	4.4.1	Fluid Type (& Category)	Gas (Category D / E)	
			Category D: Non-toxic Natural Gas	

			Category E: Flammable, and/or toxic fluids which are gases at prevailing ambient temperature and atmospheric pressure conditions and are conveyed as gases and/or liquids. Typical examples are Hydrogen, Natural Gas (not otherwise covered in Category D), Ethane, Ethylene, LPG (such as Propane & Butane), Natural Gas Liquids, Ammonia and Chlorine.	- § 5.2 Shell DEP 31.40.00.10-Gen - § 2.1 Fluid Is To Be Considered Two Phase Since No Treatment Will Be Done Before Injecting Into The Line
	4.4.2	Applicable Design Code	ANSI / ASME 31.8 - Pipeline & Field Manifolds ANSI / ASME 31.3 – PLEM and slug catcher	Applicable Design Code for Category D and E fluids, ref: Shell DEP
	4.4.3	Specification Breaks	ANSI / ASME 31.3 - at the Gas Plant Fence, as PLEM and slug catcher are within the Gas Plant Fence; as well as at the Manifold Platforms	31.40.00.10-Gen ISO 13623:2009 / BS EN 14161:200
5	Pipeline Design			
	5.1	Design Provisions	Pipelines, Node M/Fs, PLEM and slug catcher	
	5.1.1	Features	Coated Linepipe / Field Joints Coating - for external corrosion protection	- 3 Layers PE (Poly Ethylene) Main Line Coating - GTS 65 Field Joints Coating
			Concrete Coating of River Crossing Sections to Combat Buoyancy	
			Pig Launcher / Receiver with Capability for Intelligent Pigging	
			Provision for Chemical Injection Facilities on the Manifold Platforms - for Corrosion Inhibition Administration to Mitigate Internal Corrosion	

		Pipeline End Manifolds on Manifold Platforms at Gas Gathering Nodes	
		Provision for Power Supply & Related Power Distribution Ancillaries at the Manifold Platforms that will enable powered equipment operations / automation, and data transmission for monitoring of the pipeline.	Concept selected among other options - Solar Power & Individual Power Generator options at Each Manifold
		Cathodic Protection System to Combat External Corrosion	Sacrificial Anode for temporary protection of the Pipeline Impressed Current for Permanent Protection of the Pipeline Sacrificial Anode for the Steel Structures
		Sectionalisation Valves at River Crossings / at 30 km Intervals	ASME Requirement but will not be implemented for security reasons
5.1.2	Retrofit Design	Retrofit Design on Node Manifold Platforms in the light of sequential injection	
5.1.3	Power Supply Options	Power Plant at Akabel Base to generate electricity for the base operations, and possibly export to the Pipelines Manifold Platforms. Stand-alone Solar power systems are however the selected option for the manifolds but will be revisited during FEED.	Hybrid Power Generation Solution using Solar or Generator fuelled by Methanol, or Raw Gas or Diesel
5.1.4	Control & Custody	The automated block valves will be remotely controlled and monitored from BFPC Control Centre.	

Commented [AJ108]: Will it be acceptable to Nigerian Regulatory Authorities.

Commented [H109]: Yes

		Custody transfer metering will be installed at each of the SPDC wellhead M/Fs by SPDC.	
5.1.5	Health & Safety	Design will provide for healthy and safe work environments for operation & maintenance.	
5.1.6	General Environmental Responsibility	Protective devices and containment will be installed at the facilities; mainline valves will be remotely operable where practical	
5.2	Line Sizing	To Meet Throughput, Pressure & Velocity Boundary Conditions	
5.2.1	Inlet Pressure	80 barg - 100 barg (1,160 - 1,450 psi)	Delivery Pressure Available at Custody Transfer Point (CTP)
5.2.2	Outlet Pressure	60 barg - 65 barg (870 - 942 psi)	High enough to satisfy the requirements of the Gas Plant
5.2.3	Pressure Drop	Applicable Equation: - Panhandle Equation (Long Distance, Large Diameter Pipelines)	
		Steady State Hydraulic Simulation done on HYSYS & PIPESIM	
		Dynamic Hydraulic Simulation done on OLGA	
		Line Sizing Crosscheck with Panhandle Equation	
5.2.4	Velocity Considerations	Minimum Velocity to prevent surging, and to keep line swept clear of entrained solids & liquids Maximum Velocity that will not cause Erosion, Excess Noise, or Water Hammer	
5.2.5	Velocity	5 - 20 m/s: subject to Erosional Velocity Limit	API RP 14E

Commented [AJ110]: 60-65 barg

Commented [AJ111]: There could be a case when One Methanol Unit is under maintenance and Second Unit is operating under turn-down condition, in such a scenario there should be flow in the pipeline.

Commented [H112]: It will be monitored and pigged as required

	5.2.6	Minimum Velocity	High enough to prevent surging, and to keep line swept clear of entrained solids & liquids	Too low a velocity leads to settlement of water, which may lead to bottom of pipe internal corrosion
	5.2.7	Maximum Velocity	Low enough to avoid erosion, excess noise, or water hammer	Too high a velocity can increase the overall corrosion rate and also destroy any protective scale or inhibitor films.
5.3	Pipe Wall Thickness		To Meet Design Code, Pipe Material & Grade, Fluid's Pressure, Chemical Composition, Velocity, & Velocity Boundary Conditions	
	5.3.1	Fluid Category D / E	Category D: Non-toxic Natural Gas Category E: Flammable fluids which are gases at prevailing ambient temperature and atmospheric pressure conditions and are conveyed as gases and/or liquids. Typical examples are Natural Gas (not otherwise covered in Category D), Ethane, Ethylene, LPG (such as Propane & Butane), Natural Gas Liquids.	ISO 13623:2009 Shell DEP 31.40.00.10-Gen
	5.3.2	Location Class	Class 1 / Class 2 Locations	5 - 30 residential buildings per km
	5.3.3	Design Factor	Geographical 0.6, Crossings without Casings 0.6	
	5.3.4	Material & Grade	API X70.	
5.4	Pressure Rating		To Meet ANSI Standard B16.5 / API Spec 6A	
	5.4.1	Pressure Class	ANSI 900 (Class 900)	

5.5	Maintenance Provisions			
	5.5.1	Corrosion Management	Provision will be made for corrosion inhibitor injection to mitigate corrosion of the pipeline	The Gas is potentially corrosive on account of its CO ₂ content above threshold for corrosion; and water, NGL, & particulates that are inherently present
			Provision will also be made for cleaning and in-line inspection tools, shop, and in-situ maintenance	
	5.5.2	Condition Monitoring	UT Scanning	
			Intelligent Pigging	
			Leak Detection will be installed for pipelines and facilities	

Commented [AJ113]: Where are these Instrument installed?
From where the Electric Power to these will be provided? Their Security Arrangement?

Commented [H114]: These are self-powered portable equipment used only at the manifolds

6 Pipeline Routing

6.1	Route Selection	PTS has been obtained, Site Route Survey carried out and Route Selection made. The selected route will be reviewed and re-confirmed during FEED.	
		Route Optimised in the Light of Environmental Impact / Impact to Settlements	
		Consideration given to Economic Distances / Workable Routes	
		Current Concept for BBOU / ELEP is: 24" x 6.7km from BBOU - ELEP, and 36" x 31.5km from ELEP - AKAB, running through Southern Ijaw, Nembe, & Brass LGAs	
		Alternative of 42 km predominantly offshore route, aimed at minimising communities traverse was considered, but dropped, as it did not achieve that objective, in addition to the	Further the alternative offshore route will mean higher installation, and risk of hydrate

			intelligence report that it will not be acceptable to NIWA.	formation for the lower ambient temperature
6.2	Pipeline ROW		Detailed survey has been carried out and Route Selection made. Will be reviewed for final confirmation during FEED	
	6.2.1	Width	30m - (15m for P/L, including FOC, and 15m provision for Construction equipment & activities	
	6.2.2	Route Excavation	Propose to excavate route as 1 - 1.5 m deep trench that will be backfilled	
			The alternative of open ditch (3m width X 4m depth), was considered in the light that it will be permanently flooded and will therefore pose as deterrent against vandalism, - but was dropped as that objective is uncertain to be achieved.	Further the alternative open ditch canal option will mean higher installation cost that will include additional equipment and weight (concrete) coating of linepipes
	6.2.2	Crossings	Crossings of Existing pipelines, Cables, Power Lines, Roads, and Waterways to be maintained at an angle between 60 & 90 degrees	
7	Flow Assurance			
	7.1	Essence	Process conditions should be such that upstream pressure is sufficient to deliver fluid to destination, as well as avoid precipitation & deposition of solids	
	7.2	Fluid Flow	Hydraulic Flow, on own Pressure	Hydraulics Analysis to be performed to establish range of operational

Commented [AJ115]: Geotechnical Report has been furnished to Epic. Epic to confirm whether soil is suitable for backfilling?

Commented [H116]: Suitability of material will be assessed

			parameters of the Pipeline
	7.2.1	Temp - Pressure Relationship	T & P to be maintained below equilibrium line for hydrate formation, otherwise chemical (antifreeze) injection
	7.2.2	Gas Hydrate	Need for the Fluid Traverse to be maintained outside Hydrate Formation Envelop
	7.2.3	Dew Point Control	Dew Pointing & Hydrate Handling in respect of the Gas Processing will be addressed at the Gas Plant FEED

8 Pipeline Materials Selection & Corrosion Analysis

8.1	Drivers	Internal & External Corrosion Predictions and Design Life, Fitness & Compatibility	
8.2	Essence	Specification of Materials that can withstand the physical conditions and application, as well as work together	
8.3	Situations	High Pressure Transportation of Category D/E Fluid (flammable fluid), in the pressure region of 100 barg, and temperature region of 60 °C	
8.4	Components of Concern	Obnoxious Components of Concern are: CO ₂ , H ₂ S, Mercury, and Naturally Occurring Radioactive Materials (NORM)	
	8.4.1	CO ₂	CO ₂ content is up to 2%, which for the empirically established 0.1 bar partial pressure threshold for significant CO ₂ corrosion, implies corrosion of unprotected mild CS at pressures above 5 barg
	8.4.2	H ₂ S	Nil in the gas streams, - hence no sour gas corrosion to contend with
	8.4.3	Mercury	No Mercury Present, - to cause mercury amalgamation and

DeWaard-Milliams
CO₂ Corrosion
Nomogram

			embrittlement of aluminium and other metals	
	8.4.4	NORM	No NORM (Naturally Occurring Radioactive Materials), - Radon, Radium, - which can accumulate within P/L & Process Eqpt to render them radioactive over time	
8.5	Issues & Consideration		The CO ₂ content of up to 2%, renders the gas corrosive at pressures higher than 5 barg (corresponding to the empirically established 0.1 bar partial pressure threshold for significant CO ₂ corrosion), - prompting the need for CRA materials or matching corrosion management design	Internal Corrosion Issue
	8.5.1	Corrosivity Potential of the Transported Fluid (Gas)	The Gas is potentially corrosive on account of its CO ₂ content; and water, NGL, & particulates that are inherently present	Assessments Based on: - HydroCorr Software and External Resistivity Analysis - DeWaard-Milliams CO ₂ Corrosion Nomogram
		Corrosivity Potential of the Ambient Environment	Humidity, Acidity, and Oxygen in Air, and known potentially corrosive nature of the soil in the area jointly contribute to render the ambient environment corrosive	External Corrosion Issue
	8.5.2	Internal Corrosion Design	Materials Selection - NACE Codes, Consideration for MS (WT), MS (WT)+CA, MS+Inhibitor, MS with CRA	
	8.5.3	External Corrosion	All buried and above grade pipe will be externally coated; with cathodic protection system installed for the buried pipe	

		Coating Selection and Painting - NACE Code	Main Line Coating with 3 Layers PE (polyethylene) Field Joint Coating with GTS 65
		CP Design NACE Code IP	
		T/R, Junct Boxes, Cabling, (IAC Code), Groundbed design, Electrical Separation	
8.6	Line Pipe Selection	API 5L X70 Mild Carbon Steel	Duplex Stainless Steel for the Flowlines & Field Manifold Headers on the Manifold Platforms
		High Yield Strength (a favourable trade-off with required wall thickness) that can accommodate high corrosion allowance that will be needed	The Carbon Steel option would mean more internal wall thickness plus corrosion allowance
8.7	Corrosion Allowance	Up to 3 mm, and will be supported with Corrosion Inhibition	
9	Manifold Platforms		
9.1	Concept	Integrated Field Manifold Platform for both SPDC Field Manifold Header & BFPC Bulklines. Common Integration in respect of: Power Supply Fencing for Security Custody Transfer Meter on Shell side, and check meter at BFPC side	

9.2	The Manifold Platform		Comprises Field Manifold Header, Incoming & Outgoing Bulklines and Scraper Barrels, Chemical Injection System, Electrical Transformer, & Electric Room, Field Auxiliary Room (FAR) Elevated Platforms on Water c/w Perimeter fencing Two level Platform (for compactness) – segregated b/w process systems at lower level, & non-process at upper level Lower deck 5 m above mean water level (MWL) CCTV Device to monitor unauthorised intrusion Sonar Device to monitor MWL to guide operation & accessibility NAVAID to guide navigation Hybrid Solution of Solar Power & Custom Made Gas Engine Generator Communication Systems b/w M/F & Gas Plant Control Room Capability for Motorisation of Valves Shelter for personnel from weather elements	
	9.2.1	B/L Pressure Rating	ANSI 900	ANSI 1500 for the Well Flowlines
	9.2.2	Provision for Pigging	The Pipeline Manifold equipped with Pigging Facilities (Pig Launcher / Receiver) for pigging / pipe cleaning that will be done during pre-commissioning, commissioning & decommissioning of pipeline, and routine cleaning and condition monitoring of the pipeline during its operations	

9.2.3	Valving	Valving arranged and labelled so that they can be accessed safely and efficiently; the required valve can be visually identified and distinguished without error; the information required for decision making is provided; and the physical and mental workload required remains within reasonable limits. Task demands should not exceed operator capabilities	Pipeline Valves To Be Designed As Per Asme B16.34 And Asme B16.10
9.2.4	Manifold Platform Property Boundary	IVA (Identification Valve Analysis) to ensure that all valve configurations are well designed from an operational task performance perspective, and to avoid design misfits in category 1 valves (critical). A key objective is to ensure that valves are properly categorised and prioritised at an early stage in design.	
		Valve Interlock System	
		Unrestricted Access for Valves used for emergency operations (i.e. Category 1 valves)	
		Will be Fenced / Caged to Ward off Intrusion	
9.2.5	The Pig Barrel Trap	Horizontal Pig Trap, with capability for intelligent pigging	
9.2.6	Pressure Rating	ANSI 900	
9.2.7	Design Code	ASME 31.8 for the Infield M/F Pig Traps, and ASME Section VIII for Pig Trap Enclosure E ASME 31.3 for the PLEM Piping and slug catcher	

9.2.8	Corrosion Allowance	Yes, - as raw gas contains CO ₂ at threshold concentration for Corrosion	
9.2.9	Double Block & Bleed	Two separate valves in series with a bleed (vent) point in between to allow diversion to safe location of any fluid leaking through either valve	
9.2.10	End Closure	End Closure pointing away from personnel areas and critical items of equipment, in the event of pig being ejected from the pig trap	Consideration Given To Additional Safety Measures Like Using Interlocking Key System To Operate The Pig Trap System.
			Pipeline Valves To Be Designed As Per ASME B16.34 And ASME B16.10

11 Process Safety Provision & Preliminary EIA Study

11.1	Basic Consideration	Design will provide for healthy & safe working environments for operations and maintenance; as well as accommodate the environmental and heritage resource sensitivities of the region	
11.2	Process Safety Provision	Design Provision ranges from Overpressure Protection, Fire & Gas, Product Leak Detection, and Shutdown	Provision with Safeguarding (ESD) requirement in accordance to Shell standards.
11.2.1	Pipeline Overpressure Protection System	An Over-Pressure Protection System shall be provided for the protection of the Bulklines & Manifold Platforms HIPPS shall also be considered if	

			HAZOP so dictate, and if needed will be added during detailed design	
			PSVs will be installed on the pig traps to provide additional safety especially during pigging where it pups up and closes to maintain pressure balance	
			A complete blowdown / evacuation of the Pipeline will be required, and can only be achieved through Akabel where it will be possible to install such facility	
			On the other hand, for maintenance purposes, the Spec requires sectionalisation valves to be installed at pre-determined spacings on the Pipeline in order to reduce the volume of inventory to be evacuated in case of damage to any section of the Pipeline. This will require the installation of small venting systems around the Sectionalisation Valve Stations which include of course the PLEM to vent the sections of the Pipeline requiring maintenance	
11.2.2	Fire & Gas		Fire and Gas system signals will be implemented into the Pipelines & Gas Plant safeguarding system	
			A fire & gas detection system will be installed within the control room indicating main F&G detectors/initiators / F & G loops / push button, etc.	The Fire and Gas Detection System will be designed in accordance with Shell DEP and SPDC guidelines
			F & G system will be operated from the SIS system with dedicated SIS input & outputs to final indication lamps. Horns acknowledge button, lamp test button, ESD button and reset	

			button for activated lamps will also be included on the mimic panel.	
	11.2.3	Product Leak Detection	ROW Inspection by Line Patrols	
			Hydrocarbon Sensing (of External Leaks) via Fibre Optic / Dielectric Cable	
	11.2.4	Shutdown	To be initiated as a last resort of the Process Safeguarding System	
11.3		Shutdown Philosophy	Different types of shut downs, viz: - Emergency Shutdown (ESD), Operation Shutdown (OSD), Single Train Shutdown, and Unit Shutdown, - will be designed to be initiated depending on process upsets and their hazardous levels.	The process operating, control and start-up / shutdown philosophy to be detailed at FEED
	11.3.1	Emergency Shutdown System (ESD)	The ESD will be designed to be initiated by fire, manual push buttons or by loss of instrument air supply	
	11.3.2	Operational Shutdown (OSD)	The Operational Shut Down (OSD) will cut off the fluid supply from the pipelines as well as block the gas supply to the Methanol & Power Plants, albeit that an ESD will always initiate the OSD	
	11.3.3	Single Train Shutdown	The single train shutdown will cut off all fluid supply and discharges from a single equipment train allowing the parallel equipment train(s) to continue operation	
	11.3.4	Unit Shutdown (USD)	The Unit Shutdown will stop a dedicated unit of the plant	

			On the other hand, for maintenance purposes, the Spec requires sectionalisation valves to be installed at pre-determined spacings on the Pipeline in order to reduce the volume of inventory to be evacuated in case damage to any section of the Pipeline. This will require the installation of small venting systems around the Sectionalisation Valve Stations which include the PLEM to vent the sections of the Pipeline requiring maintenance	
11.4	Hazardous Area Classification (HAC)		Will be in compliance with Shell DEP 80.00.10.10-Gen in cross reference with API RP 505 and IP 15	Shell DEP 80.00.10.10-Gen
	11.4.1	Zones Classification	Zone 1 - where a flammable atmosphere is continuously present for long periods Zone 2 - where a flammable atmosphere is likely to occur in normal operation Zone 3 - where a flammable atmosphere is not likely to occur in normal operation and if it occurs it will exist only for short period	
	11.4.2	Extent of Zone	The standard 6m Radius will be used as general guide in this Basic Design scope, while specifics & details will be covered at FEED	
	11.4.3	Location of Equipment	Will be done with the consideration to avoid ignition of flammable hydrocarbons by minimising the probability of coincidence of a flammable atmosphere and a source of ignition.	

		In general will be more than 3m away from the launcher and receiver barrels which are the major points of release on the manifold.	
11.5	Preliminary EIA Study	An environmental impact assessment is currently in progress for the gas plant site, and will also be carried out for the Pipeline Route and associated Manifold Platforms	
		Will cover the entire Plant Complex (Gas Plant, Power Plant, and the Methanol Plant), the Pipeline Route, and the associated Manifold Platforms	
		Only the phase-1 Pipeline Route (BBOU - ELEP - AKAB) will be covered now. The other routes will be covered later, close to their on-stream dates, otherwise we will end up with obsolete EIA when their times come	
12	Power Generation	IPP as an option at the Gas Plant to provide the power needs of the Gas Plant, , and the Methanol Plant Complex, and possibly the Pipelines and manifolds Platforms systems.	Full Specification to be detailed by the IPP Scope. Preferred power supply option for the Pipelines and Manifolds to be revisited during FEED.
12.1	Primary Power System	Gas Turbines	N+1 philosophy, and to be synchronised
		Power Generation at 11 kV, and stepped down to 6.6 kV and 415 V for high voltage and low voltage supply to the Gas Plant and Manifold Platforms as required	Emergency → essential to safe start up / shutdown
12.2	Secondary Power System	Gen Set	Emergency → essential to safe

			start up / shutdown
		Power requirement and classification - normal or emergency depending on equipment specification.	Reference with Load List Vital & Essential Loads to be defined & agreed
	12.2.1	Uninterruptible Power System	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load
	12.2.2	Solar Power Option for Emergency	Option for Emergency Back-up with Solar Power will be considered during FEED
13	Power Supply		
13.1	Power Supply to the Remote Manifold		Standalone solar power system is the selected option for the manifolds. This option will be revisited during FEED. Underground Power Cable from the power plant at Akabel is one of the options that will be considered during FEED
	13.1.1	Power Supply Options A	Six main options were evaluated, from which solar power was selected. This will be revisited during FEED. The Motorization of Valve shall be considered for Valves above 12"
		1. Underground Power Cable at all manifolds	Buried underground cables
		2. Overhead Power Cable at all manifolds	Power cables run through transmission towers
		3. Composite Power Cable (with FOC) at all manifolds	Composite cable with power and data transmission capability.

Commented [ME117]: 1.EPIC study in 2017 recommended Solar Power in all locations.
2.TATA study in 2021 recommended underground cable at ELEP and Diesel Gen at BBOU. Other locations deferred until later.

13.1.2	Power Supply Options B	4. Solar power at all manifolds – selected option	Photo-voltaic cells and power bank arrays with inverters
		5. Combination of Cable and Solar Power	Part underground and part photo-voltaic cell array
		6. IPP in all manifolds	Gas engines, thermo-electric generators and micro-turbine generators
		For Hydraulic Valve operation, two main supply options were evaluated as Option B.	
		1. No Power to Manifolds	
13.1.3	Distribution of the Cable Power as an alternative to the selected Solar power option	2. No Power to Manifolds except Independent Solar Power	Photo-voltaic cells and power bank arrays with inverters
		Underground power cable buried along bulk pipeline From the Power Plant at Akabel to the Remote Manifold Platforms at ELEP, BBOU, NUNR, SEIB, OPUG and OKPO	Cable sizing with reference to Load requirement, ampacity, voltage drop and short-circuit current.
		Underground power cable buried along bulk pipeline From the Power Plant at Akabel to the AKAB PLEM	
		Atex (Ex) / NEMA Junction boxes provided along the MF for cable hook-up.	
13.1.4	Emergency & Essential Services Power	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	vital load → needed to run / be available during emergency s/d for

	Systems		safety of personnel
		Power requirement and classification - normal or emergency depending on equipment specification.	Reference with Load List Vital & Essential Loads to be defined & agreed
13.1.5	Uninterruptible Power System	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	The Vital & Essential Loads to be defined & agreed
13.1.6	Solar Power Option for Emergency	Option for Emergency Back-up with Solar Power will be considered during FEED	
13.1.7	MCC	Cubicle provision for supply of field MFs dependent on load requirement.	Cubicle provision and sizing dependent on load requirement
13.1.8	Earthing & Equipotential bonding	Earthing of MFs to desired earth resistance $\leq 4\Omega$	SHELL DEP 33.64.10.33-GEN
13.1.9	Lighting, small power and welding receptacle	Minimum illumination provision (10 - 20 lux) photocell controlled on MFs with Atex (EX) / NEMA lighting fixture. Small power provision for portable hand tools and welding socket provision	Will be addresses during FEED.
		Perimeter and area lighting for facility location and manifold operation area. Small power welding receptacles to be provided with the hazardous area classification as a consideration.	Welding during shutdown TAM only
13.1.10	Single Line Diagrams	Overall and individual single line diagrams to show flow of power supply from the power generation equipment to the required loads.	

	13.1.11	Load List	Detailed load list of each M/F based on equipment that require power supply, area and fence lighting.	To determine the load requirement for each M/F for adequate sizing of generation equipment.
	13.1.12	Hazardous Area Classification	Will be in compliance with Shell DEP 80.00.10.10-Gen in cross reference with API RP 505 and IP 15	Shell DEP 80.00.10.10-Gen
	13.1.13	Foot prints	Detail the extents of the selected option so as to have an estimated Manifold Platform size	
13.2	Power Supply to the Gas Plant			
	13.2.1	Power Supply Options	Step Down of IPP Voltage from 11 kV to 6.6 kV & 415 V for Power Supply to the Gas Plant	
	13.2.2	Distribution of the Cable Power	Underground power cable from the Power Plant to the Gas Plant	Cable sizing with reference to Load requirement, ampacity, voltage drop and short-circuit current.
			Atex (Ex) / NEMA Junction boxes provided along the MF for cable hook-up.	
	13.2.3	Emergency & Essential Services Power Systems	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	vital load → needed to run / be available during emergency s/d for safety of personnel
			Power requirement and classification - normal or emergency depending on equipment specification.	Reference with Load List Vital & Essential Loads to be defined & agreed
	13.2.4	Uninterruptible Power	UPS supply via dedicated panel connected to emergency bus	The Vital & Essential Loads to

	System	(switchboard) to supply vital & essential load	be defined & agreed
13.2.5	MCC	Gas Plant Scope / Co-Design with MCC Vendor	
13.2.6	Earthing & Equipotential bonding	Earthing of Plant Skids to desired earth resistance $\leq 4\Omega$	Shell DEP 33.64.10.33-Gen
13.2.7	Lighting, small power and welding receptacle	Minimum illumination provision (10 - 20 lux) photocell controlled on Plant Units with Atex (EX) / NEMA lighting fixture. Small power provision for portable hand tools and welding socket provision	Will be addressed during FEED.
		Perimeter and area lighting for facility operation area. Small power welding receptacles to be provided with the hazardous area classification as a consideration.	Welding during shutdown TAM only
13.2.8	Load List and Single Line Diagram	This Load List will be addressed The Single Line Diagram shall be generated	
13.2.9	Hazardous Area Classification	Will be addresses at FEED, and in accordance with Shell DEP 80.00.10.10-Gen in cross reference with API RP 505 and IP 15	
		The location of equipment to be more than 3m away from the launcher and receiver barrels of the Plant's PLEM, and Process Vessels which are the major points of release of Hydrocarbon.	Intent is to minimise the probability of coincidence of a flammable atmosphere and a source of ignition.

APPENDICES

1. Dashboard of Schematics, Production Forecast, Fluid Properties & Composition, and Design Throughputs
2. Workbook of Pipeline Route Data, Production Forecast, & Fluid Composition

Appendix-1: Dashboard of Schematics, Production Forecast, Fluid Properties & Composition, and Design Throughputs

Appendix 1	Dashboard of Schematics, Production Forecast, Fluid Properties & Composition, and Design Throughputs	 Dashboard of Schematics, Productic
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Appendix-2: Workbook of Pipeline Route Data, Production Forecast, & Fluid Composition

Appendix 2	Workbook of Pipeline Route Data, Production Forecast & Fluid Composition	 Workbook of Pipeline Route Data, Productio
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